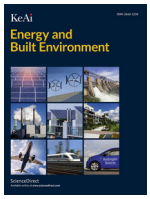
Energy and Built Environment 6 (2025) 248–276

Contents lists available at ScienceDirect

Energy and Built Environment

journal homepage: http://www.keaipublishing.com/en/journals/energy-and-built-environment/



Full Length Article

Advancements in cooling techniques for enhanced efficiency of solar photovoltaic panels: A detailed comprehensive review and innovative classification

Mohamad Abou Akrouch [a,b], Khaled Chahine [c], Jalal Faraj [a,d], Farouk Hachem [a], Cathy Castelain [b], Mahmoud Khaled [a,d,*]

[a] Energy and Thermo-fluid Group, Lebanese International University, LIU, Bekaa, Lebanon
[b] Laboratory of Thermal and Energy of Nantes LTEN, University of Nantes, Nantes, France
[c] College of Engineering and Technology, American University of the Middle East, Kuwait
[d] Energy and Thermo-fluid Group, The International University of Beirut, BIU, Beirut, Lebanon

ARTICLE INFO

Keywords:
Comprehensive
Comparative
Review
Photovoltaic panel
Cooling techniques

ABSTRACT

Solar photovoltaic (PV) cells have emerged as the primary technology for producing green electricity. This innovation harnesses direct sunlight to generate power and its flexibility of installation has drawn significant investment in PV panels. Despite numerous benefits, these cells are hindered by a decline in efficiency caused by elevated cell temperature. As such, researchers have undertaken extensive investigations into possible solutions aimed at enhancing the performance of photovoltaic cells using diverse techniques. This review paper provides a thorough analysis of cooling techniques for photovoltaic panels. It encompasses both passive and active cooling methods, including water and air cooling, phase-change materials, and various diverse approaches. Within each category, it delves into detailed sub-categories, such as evaporative cooling, water immersion, floating systems, water pipes, cooling channels, water sprayers, jet impingement, geothermal cooling, and natural convection enhanced by PV designs. It also covers forced convection using cooling ducts, heat sinks, and air collectors, alongside the integration of Phase Change Materials (PCMs), nanofluids, radiative cooling, thermoelectric methods, heat pipes, heat pumps, and other innovative techniques. Each of these approaches is illustrated with specific schematics and thoroughly discussed and compared. Furthermore, this paper introduces an original classification system for these cooling methods applied to photovoltaic panels, offering valuable guidance for future research and insights into improving efficiency.

1. Introduction

The steady growth of population and economic activity has triggered an unprecedented surge in energy demand, encompassing diverse sectors. Consequently, the extensive exploitation of non-renewable fossil fuels has contributed to their depletion while simultaneously elevating both expenses and carbon dioxide emissions in the atmosphere [1]. Thus, there is a pressing need to investigate and advance alternative sources of renewable energy that can substitute for conventional fossil fuels [2].

Investing in solar energy is a viable option to capitalize on the abundant availability of high radiation from the sun worldwide. Photovoltaic (PV) panels are particularly advantageous due to their uncomplicated mechanism that utilizes semiconductor materials to convert sunlight into electrical power. This technology has advanced significantly in recent years, establishing itself as a prominent renewable energy source with notable benefits such as diminished carbon dioxide emissions, rapid return on investment, versatility across various fields, and resilience under harsh conditions. Unfortunately, this technology exhibits a comparably diminished degree of conversion efficiency due to numerous factors that exert influence [3], including but not limited to dust accumulation [4], reflection [5], angle of inclination variation [6,9], orientation [7], shading effects [8], radiation exposure [5], and temperature [9]. Temperature bears the greatest impact among these parameters as solely visible light is transformed into electrical energy by photovoltaic cells while other spectral regions such as infrared convert into heat energy, which diminishes their overall effectiveness.

Given the potential benefits of improved energy efficiency, cost reduction, and environmental preservation linked to advancements in photovoltaic cell performance, researchers have been actively investigating diverse cooling methods to enhance the effectiveness and cost-efficiency of photovoltaic panels. This surge in interest has generated

* Corresponding author.
E-mail address: mahmoud.khaled@liu.edu.lb (M. Khaled).

https://doi.org/10.1016/j.enbenv.2023.11.002
Received 3 September 2023; Received in revised form 20 October 2023; Accepted 3 November 2023
Available online 4 November 2023
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a wealth of studies delving into photovoltaic cooling techniques. Going forward, the need for comprehensive review papers summarizing recent cooling strategies will be paramount to drive further advancements in this field. Some of these papers have focused on specific methods, such as the work by Ali, H. M. [10] on phase change materials, Suresh, Ashij K., et al. [11] on nanofluids as coolants, Bhakre, S. S. [12] on water cooling, Preet, Sajan [13] on water and PCM cooling, Mohd Yusof Othman et al. [14] on air cooling, Kane, Aarti et al. [15] on thermoelectric cooling, Elbreki, A.M. et al. [16] on passive cooling, Bahaidarah, Haitham M.S. et al. [17] on uniform cooling, and Siah Chehreh Ghadikolaei, S. [18] offered a comprehensive review of cooling techniques, but without focusing on thermoelectric and evaporative cooling. Similarly, Verma, Shrey, et al. [19] presented an extensive review, but without emphasis on water and evaporative cooling techniques.

This study seeks to provide a comprehensive and up-to-date review of advanced cooling techniques for photovoltaic systems, highlighting their pivotal role in increasing the sustainability and efficiency of these systems. To achieve this goal, the review begins with a brief theoretical foundation on PV technology and its critical influencing factors, emphasizing the paramount importance of cooling technologies in improving overall system performance. The investigation then explores different perspectives and classifications of advanced cooling techniques, including passive and active water cooling, free convection and forced air, phase change materials acting as a thermal regulator, and other innovative approaches. The review neatly organizes a wide range of cooling techniques in a clear and concise manner, complemented by explanatory diagrams that illustrate the principles of each technique. Furthermore, a comprehensive analysis, discussion, and comparison of these techniques based on their respective improvements in performance are conducted to provide key recommendations for future research and development in this field. The originality of this study lies in its systematic approach to the classification and evaluation of different cooling techniques, which constitute a significant contribution to the development of PV cooling techniques.

2. Theoretical background

The phenomenon of the photoelectric effect was first discovered by French scientist Edmond Becquerel, who noticed that some materials produce a certain amount of electricity when exposed to sunlight. This discovery was refined and developed by researchers over time, leading to the development of modern photovoltaic cells available today. These cells typically have three layers: an upper layer with extra electrons (n-type), a bottom layer with fewer electrons (p-type), and a middle layer with slightly fewer electrons, creating a p-n junction. When sunlight hits the cell's top surface, it knocks an electron from a silicon atom, leaving a positive region or "hole." The released electron moves toward the upper n-type layer, which accepts electrons, while the hole moves to the lower p-type layer, which accommodates holes. This process continues as long as there's sunlight. Connecting a wire between the top and bottom creates a path for electrons to move, generating electric current for devices. Fig. 1 summarizes the basic idea of how a solar cell works.

2.1. The basic governing equations

The equations used to calculate the efficiency of the photovoltaic panel are shown below:

➢ The electrical efficiency equation is defined as:

$$\eta_{(electrical)} = \frac{P_{\text{output}}}{P_{\text{input}}} \times 100 \tag{2.1}$$

where:

- P_{output}: Electrical power in Watt
- P_{input}: Solar received Power in Watt

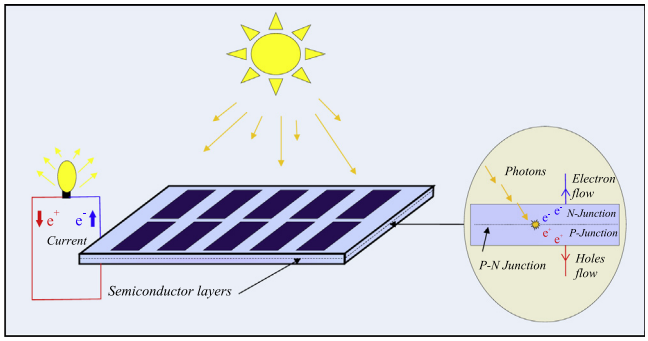


Fig. 1. Basic operating principle of a solar cell [20].

➢ The percentage (%) of enhancement of electrical efficiency (%) is calculated as follows:

$$\eta\% = \frac{\eta\%\ (Cooled\ PV) - \eta\%\ (Standard\ PV)}{\eta\%\ (Standard\ PV)} \times 100 \tag{2.2}$$

where:

- $\eta\%$ represents the percentage improvement in electrical efficiency.
- $\eta\%$ Cooled PV is the efficiency of the cooled photovoltaic system.
- $\eta\%$ Standard PV is the efficiency of the standard-uncooled photovoltaic system.

➢ The output power is defined as:

$$P_{\text{output}} = V \times I \tag{2.3}$$

where:

- I: Current in Ampere
- V: Voltage in volts

➢ The thermal equation for PV panels is defined as:

$$\eta_{(thermal)} = \frac{\dot{m} c_p \left(T_{water\ out} - T_{Water\ in}\right)}{P_{\text{input}}} \times 100 \tag{2.4}$$

➢ The total efficiency equation is defined as follows:

$$\eta_{(total)} = \eta_{(electrical)} + \eta_{(thermal)} \tag{2.5}$$

2.2. Parameters affecting photovoltaic cell efficiency

Many influencing parameters affect the efficiency of photovoltaic panels and reduce their output power, which is mainly related to the type of PV technology, ambient conditions, system equipment, and system quality of grid-connected PV systems. In addition, each of these parameters has several influencing sub-parameters, as shown in Fig. 2.

Among these parameters, temperature predominates in efficiency degradation, owing to the natural properties of photovoltaic materials that convert sunlight into electricity. The photovoltaic cell uses between 700 and 1100 nm solar spectrum to produce electrical energy (see Fig. 3), whereas other wavelengths are either reflected or passed through the panel and converted into heat, thus increasing the temperature of the solar cell above the normal operating temperature.

More precisely, an increase in temperature greater than 25 °C [22] causes efficiency degradation of approximately 0.5 % to 0.6 % for each 1-degree temperature increase depending on the type of solar cell material [23,24] and a typical PV module converts only 6–20 % of the incident solar radiation into electricity [25,26]. From this perspective, the importance of photovoltaic cooling has emerged and has become a race among researchers.

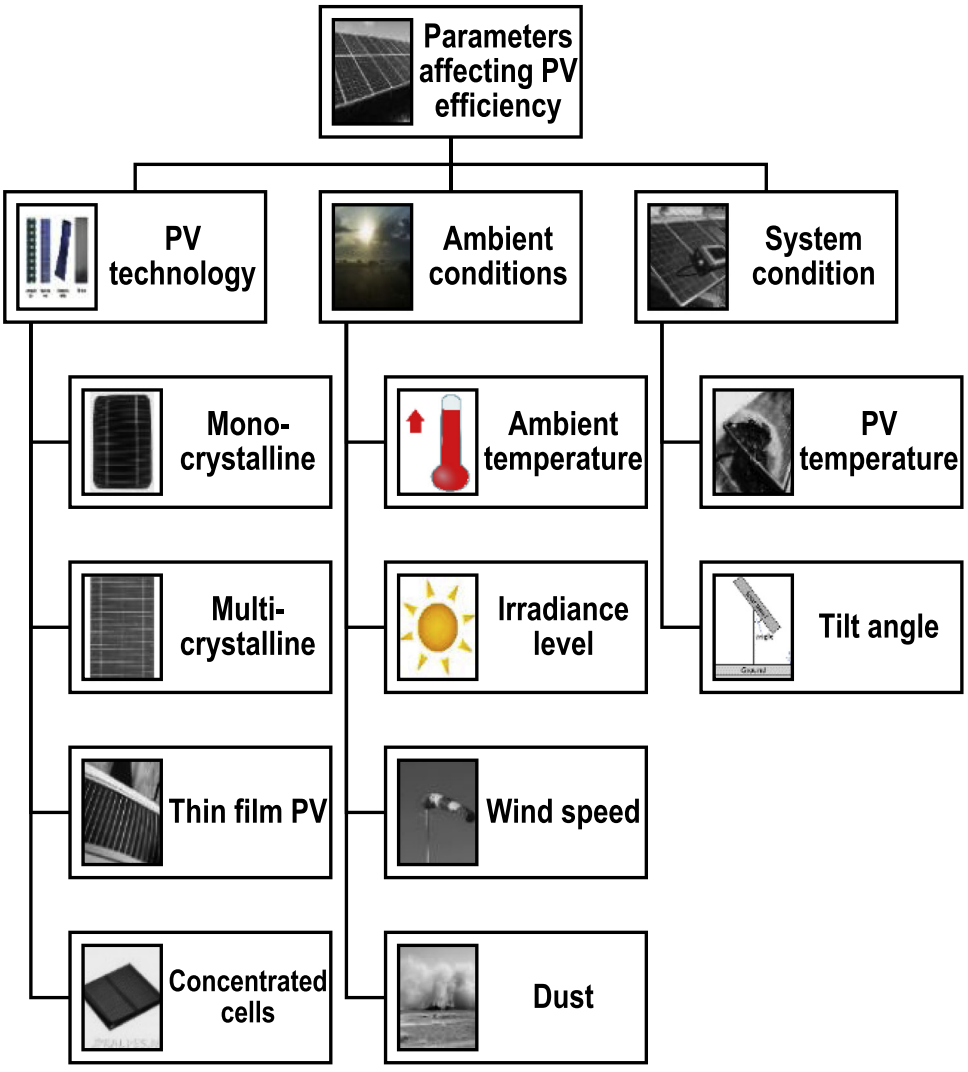


Fig. 2. Parameters affecting PV panel efficiency.

3. Main cooling techniques

Photovoltaic panels convert sunlight into electricity, but they become less efficient as they get hotter. To tackle this, there are different cooling methods, as mentioned in Fig. 4:

1. Water Cooling: The water is circulated to absorb and dissipate heat, keeping the panels with high efficiency.
2. Air Cooling: Fans or convection remove heat, a more straightforward method but less effective at higher temperatures
3. PCM Cooling: Phase change materials store and release heat to stabilize performance.
4. Innovative Techniques: New approaches like coatings, nanomaterials, thermoelectric, or heat pipes enhance heat dissipation.

All of these methods, with their types, aim to keep panels at the right temperature, making them work better, last longer, and help renewable energy progress.

In addition, these techniques can be categorized into passive and active cooling techniques, as shown in Fig. 5. The passive cooling technique is characterized by natural flow without using any external power, whereas active cooling is characterized by the use of mechanical assistance (such as pumps, fans, suction devices, etc.) to increase the contact flow with the coolant and increase heat transfer.

Passive cooling is an effective method that utilizes natural water flow, eliminating the need for pumps to cool photovoltaic panels. However, its cooling capacity is limited, and excess heat in the water must be managed. In contrast, active cooling involves forced water flow using pumps to regulate panel temperature by adjusting water speed. However increasing the flow rate requires more pumping power, raising energy consumption. Similarly, natural convection occurs as hot and cold gas exchange heat due to buoyancy. For instance, when a panel heats up, it warms the air around it, reducing air density. This causes heated air to rise, creating a natural convection current. The enhancements of natural convection include using fins to expand the heat-transfer area. Conversely, forced convection relies on airflow across panels' front and back to maintain standard temperatures, which requires a significant amount of fan power.

3.1. Water cooling methods

The excellent heat absorption properties of water make water-cooling a specialized technique for improving the performance of photovoltaic systems. By efficiently dissipating excess heat, this approach contributes to improved temperature control and overall PV system efficiency. It can be implemented as either passive or active cooling, providing adaptable solutions to meet specific requirements.

3.1.1. Water immersed PV

Immersed photovoltaic systems offer an effective way to enhance solar power generation. This passive cooling technique involves immers-

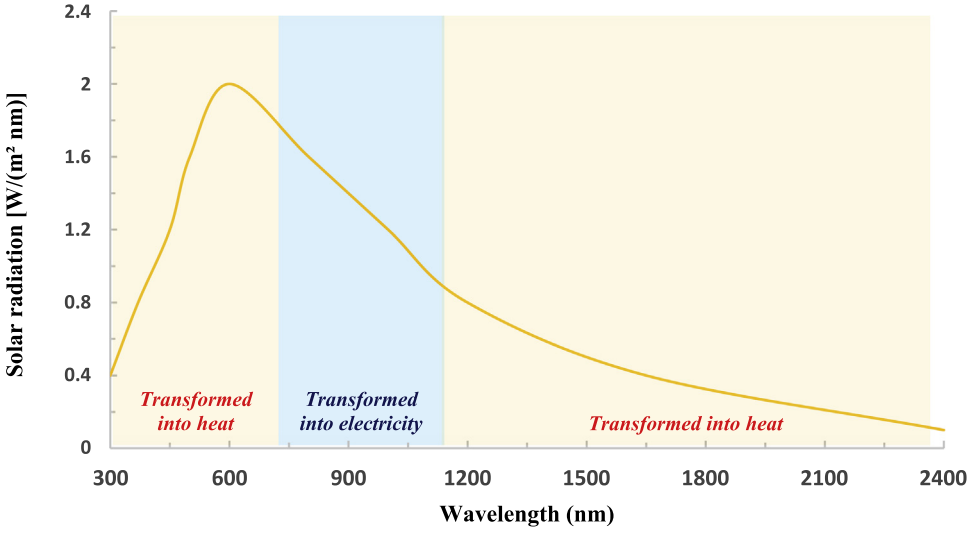


Fig. 3. Solar spectrum received by solar cells [20–21].

ing PV panels directly into a water tank at a specified depth, as shown in Fig. 6. By harnessing the cooling properties of water, this approach efficiently dissipates heat and maintains controlled operating temperatures, which can lead to improved PV system performance.

Saurabh Mehrotra et al. [27] have studied the performance of a solar panel with a water immersion cooling technique to maintain its surface temperature and provide better efficiency at extreme temperatures (see Fig. 6). The results showed that the panel efficiency increased by about 17.8 % at a water depth of 1 cm. Similarly, Lanzafame et al. [28] investigated the electrical and thermal performances of a single submerged photovoltaic solar panel system at different depths. The results show that PV panels have been found to work successfully when they are submerged in shallow water, with an increase in photovoltaic efficiency by approximately 15 % at a water depth of 4 cm.

3.1.2. Evaporative cooling

Evaporative cooling is a practical technique for optimizing photovoltaic systems. By using water evaporation to lower the air temperature and maintain a comfortable environment (as shown in Fig. 7), this technique proves effective in cooling photovoltaic cells and enhancing their performance (Table 1).

Evaporative methods mainly belong to the category of passive techniques and can be implemented using different materials to simulate vapor transmission. Materials such as cotton wicks, heat spreaders, and bionic vaporizers can be used to simulate vapor. These approaches demonstrate the effectiveness and ability to reduce PV cell temperatures by up to 20 °C. However, the application of this cooling methodology using nanofluids was ineffective.

3.1.3. Floating PV

Floating cooling techniques offer a unique solution for optimizing photovoltaic systems. By placing photovoltaic panels on water surfaces, these methods take advantage of the cooling effect of water to dissipate heat efficiently and improve temperature control. This approach also allows large-scale floating solar power plants to be built, using unused areas such as rivers, lakes, and seas, while still maintaining higher unit efficiencies due to the benefits of water cooling. Torunn Kjeldstad et al. [40] studied the effect of floating cooling on the performance of the photovoltaic string. The results showed that the PV string in direct contact with water had an average of 5–6 % higher than that of the air-cooled string.

3.1.4. Water pipes and cooling channel techniques

Water pipe and cooling channel techniques are practical cooling strategies for photovoltaic systems. These methods effectively dissipate heat from the PV panels, which contributes to efficient temperature control and potential performance improvement. S. Kianifard et al. [41] studied the performance of a serpentine half tube for PV/T enhancement (see Fig. 8). The results showed that the thermal efficiency of the proposed model increased by approximately 10 to 13 % and the electrical efficiency by approximately 0.4 to 0.6 % (corresponding to percentage improvements between 3.6 % to 5.5 %) compared to the traditional models (Table 2).

Cooling tubes can lower PV panel temperatures by 10–25 °C and increase electricity production efficiency by over 13 %. Effectiveness depends on the materials used and various tube designs (full, half, finned) that can be configured in arrangements such as serpentine, linear, and circular. This approach can be complemented by various cooling technologies, including fluid-based options (air, water, nanofluids) and the incorporation of phase change materials. However, it is important to note that most of these methods fall under the category of active cooling technologies.

3.1.5. Photovoltaic thermal

The photovoltaic thermal collector, shown in Fig. 9, operates as a hybrid system that not only generates electricity but also satisfies the heating power requirements while effectively controlling the operating temperature of the photovoltaics (Table 3).

The PV/T system enhances electrical power generation and harnesses additional thermal energy. However, this technology is typically classified as an active cooling method that relies on water pumping power. Its performance is influenced by diverse cooling channel designs, including porous, converging, double channel, and various designs. Implementing this technology carries the potential to improve the average efficiency by more than 30 %.

3.1.6. Earth–water heat exchanger

Groundwater cooling provides a practical approach to optimizing photovoltaic systems. By taking advantage of the temperature difference between the groundwater and the surrounding air, this method efficiently dissipates heat from the PV panels, contributing to better temperature management and potential performance improvement. Rahim Jafari [63] uses optimized geothermal heat exchangers (see Fig. 10) to cool the photovoltaic panels through a back-side micro-channel heat sink. The results showed a 9.8 % improvement in the average net electricity generation compared with the uncooled unit.

Sanjeev Jakhar et al. [64] experimentally and theoretically studied the performance of PV/T systems integrated with glazed tubes and earth water heat exchanger cooling. The results showed that the temperature

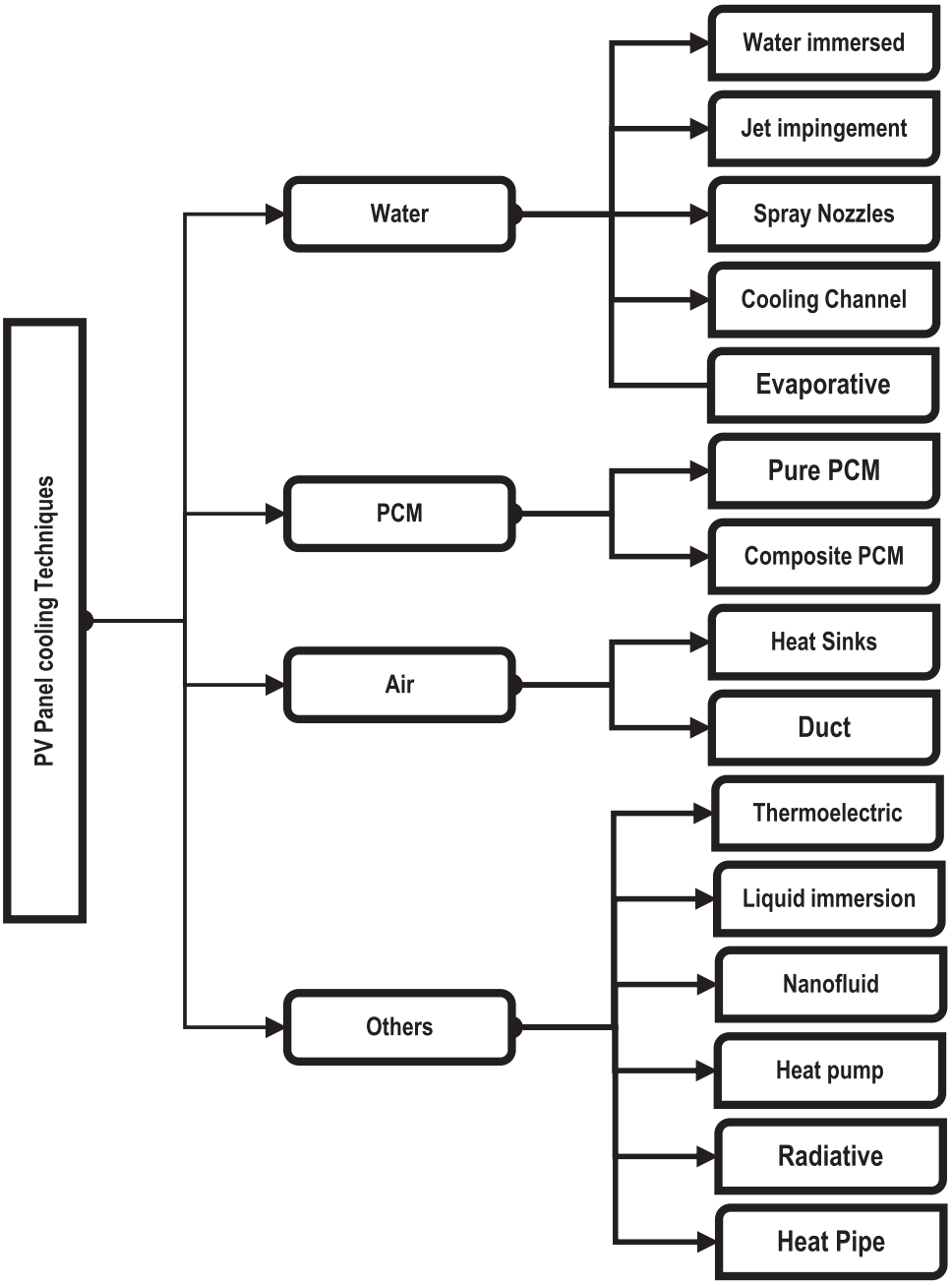


Fig. 4. Classification of the main cooling techniques.

of the uncooled module was 73 ˚C, while the temperature of the cooling module was between 43.68–49.64 ˚C, and the electrical and thermal efficiencies were in the range of 8.26–8.52 % and 44.06–55.45 % respectively. A similar numerical study by Ruoping et al. [65] evaluated the performance of photovoltaics cooled by geothermal exchangers in a dry climate. The results showed a temperature decrease of 26.8 % on average daily, which resulted in an annual electricity yield of 7.9 % compared with conventional PV systems. In addition, Jakhar et al. [66] studied the performance of an integrated photovoltaic thermal solar (IPVTS) system with an earth–water heat exchanger (EWHE). The results showed that with EWHE cooling, there is an increase in the experimental electrical efficiency of the IPVTS by 1.02–1.41 % (corresponding to percentage improvements between 11 % and 16.5 %) compared to the non-cooled condition.

3.1.7. Spray nozzles

Spray nozzle cooling provides an effective way to enhance photovoltaic systems. Through the use of directed sprays of water, this technique efficiently dissipates heat from the PV panels, contributing to efficient temperature management and potential performance improvement. S. Nizetic et al. [67] experimentally examined the performance of photovoltaic panels using a water-spray cooling technique (see Fig. 11) applied to the front, back, or both simultaneously. The results showed that the highest cooling performance was obtained by simultaneous cooling, with a maximum total increase of 16.3 % (effective 7.7 %) in electrical energy production and a total increase of 14.1 % (effective 5.9 %) in the electrical efficiency of the photovoltaic panel. Furthermore, the panel temperature can be reduced from an average of 54 ˚C (uncooled PV panel) to 24 ˚C (Table 4).

The effectiveness of a water spray cooling method depends primarily on how the water cooling system is structured, including factors such as

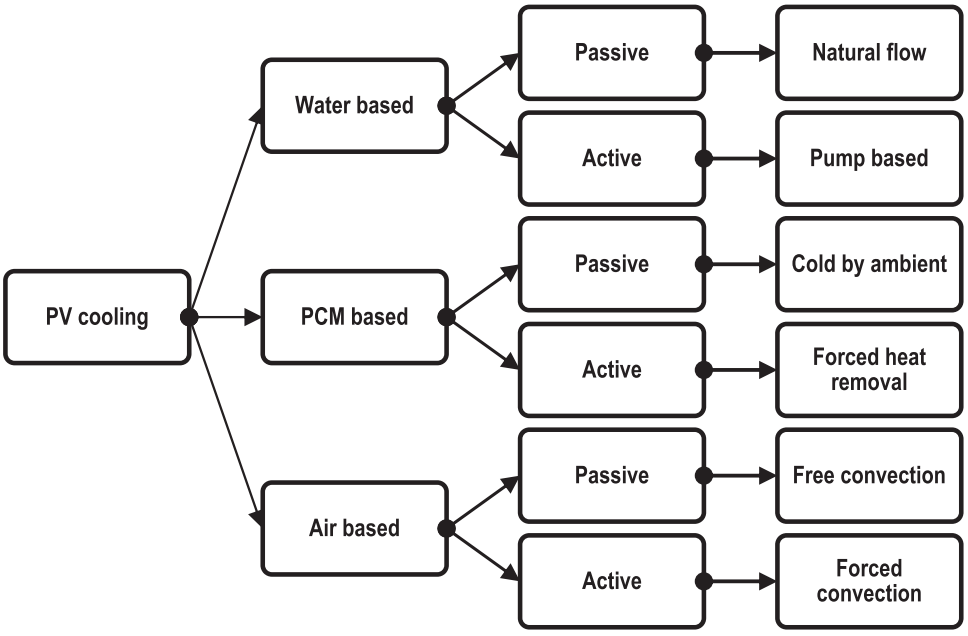


Fig. 5. Active and passive classification of PV cooling techniques.

Table 1
Summary of most experimental studies on photovoltaic panels with different types of evaporative cooling techniques.

Authors	Type of cooling	Cooling technique	Temperature results	Electrical Results
Mustapha Dida et al. [29]	passive	Burlap that was attached directly to the back surface of the module	Temperature decreased by 20 °C	η_{el} increased by 14.75 %
Evelyn Drabiniok et al. [30]	passive	Design a bionic evaporation foil	Temperature decreased by up to 11.7 °C	——
Chandrasekar et al. [31]	passive	The cotton wick in combination with water, Al2O3/water nanofluid, and CuO/water nanofluid	——	Cotton wick improved η_{el} by 1.4 % (corresponding to percentage improvements of 15.5 %), while adding nanofluids reduced the efficiency
Abdul Hai Alami et al. [32]	passive	Adding a synthetic clay layer at the back of the module to evaporate a thin film of water	——	A max increase of 19.1 % in the output power
Chandrasekar et al. [33]	passive	Heat spreader with cotton wick	Temp. reduced by 12 %	Electrical yield increased by 14 %
Morteza Ebrahimi et al. [34]	Active	Natural vapor facing back side using natural vapor simulator, at a flow rate between 1.6 to 5 g/min	Temp. decreased from 7 to 16 °C	η_{el} increased by 12.12 % to 22.9 %
Abdul Hai Alami [35]	Passive	Synthetic clay and aluminum as different alternative backing materials than standard Tedlar	Temp. reduced from over 28 °C to about 10 °C	Boosted energy production by 34.6 %
Lucas et al. [36]	Passive	Evaporative chimney	——	η_{el} improved by an average of 4.9 % to a maximum of 7.6 %
Haidar et al. [37]	passive	Water was supplied to the back of the PV panel from a tank by gravity	Temp. decreased by more than 20 °C	η_{el} increased by about 14 %
Ephraim Bonah Agyekum et al. [38]	passive	Cotton wick mesh that absorbs water from a perforated tube	Temperature decreased by 23.55 °C	Power improved by 30.3 %, and η_{el} improved by 1.53 % (corresponding to percentage improvements of 11.9 %)
Almuwailhi et al. [39]	Passive & active	Insulated duct under natural or forced convection	——	η_{el} increased by 1.2 %, 4 %, 2.7 %, and 3.8 % under natural convection, forced convection, natural evaporative, and forced evaporative, respectively.

the variety of nozzles used, pipe diameter, water flow rate, and irrigation method, which can be automatic, continuous, or pulsating spray. This approach shows a dual advantage by reducing the PV temperature while ensuring a consistent clean surface of the PV panel. As a result, it has the potential to lower the PV temperature by up to 25 °C and improve electrical efficiency by over 17 %. However, this technique is classified as an active cooling method.

3.1.8. Microchannel cooling techniques

Microchannel water cooling offers a sophisticated way to improve the performance of a photovoltaic system. By utilizing micro-sized channels to enhance heat dissipation, this technique aims to efficiently manage temperatures and potentially raise the overall performance of the PV system. Peyvand Valeh-e-Sheyda et al. [80] studied the performance of two-phase flows (air and water) in a small hybrid microchannel solar

Table 2
Summary of most studies on photovoltaic panels with water pipes and cooling channels.

Authors	Type of study	Type of cooling	Cooling technique	Thermal results	Electrical Results
Bhaidara et al. [42]	Num. & Exp.	Active	Water cooling systems installed on the back surface of the PV panel	Temperature reduced to about 20 %	η_{el} increased by 9 %
Shrinivas Bojanampati et al. [43]	Exp.	Active	Using forced air and water-cooling	——	The output power improved by about 10 % with forced air cooling. While increased by 48 % with water-cooled modules
İlhan CEYLAN et al. [44]	Exp.	Active	A cooling system consisting of a simple tube placed on a PV module	——	η_{el} improved from 10 % to 13 %
Natale Arcuri et al. [45]	Exp.	Active	Water cooling system and air cooling on the back of the panels	A max temp. change is 4 K achieved using 4 water channels	Annual η_{el} with water-cooling is 12.65 %, and with air-cooling is 12.58 %
Rahman et al. [3]	Num. & Exp.	Active	Water cooling system	Temperature decreased from 56 ˚C to 22.4 ˚C	At 1000 w/m2 η_{el} increased by 1.23 % (corresponding to percentage improvements of 27.3 %)
McColl et al. [46]	Exp.	Active	Water cooling and sun tracking system	At an ambient air-water temperature	Electrical energy production increased by 22 %. In addition, the Annual yield increased by 18 % using water cooling, and only 15 % using air cooling
Zhijun Peng et al. [47]	Exp.	Active	Water cooling system	——	The efficiency of solar PV could increase by an average percentage of 47 % under cooling conditions.
Mohammad Mafizur Rahman et al. [4]	Exp.	Active	Finned tube heat exchanger attached to the back of the photovoltaic panel	Temp. reduced by 10.28 ˚C	——
Kabeel et al. [48]	Exp.	Active	Reflectors and cooling methods	——	Power improved by 16.81 %, 21.62 %, 35.13 %, & 39.69 % using the reflector alone, reflector and air-cooled, reflector and water-cooled, and reflector with both air and water-cooled, respectively.
Shmroukh [49]	Exp.	Active	Various types of water-cooling systems	——	The open-loop water cooling system increased the efficiency by 5.1 % (corresponding to percentage improvements of 82 %)
Madhu Sharma et al. [50]	Exp.	Passive	Designed and developed a special pass reverse flow absorber collector	——	For every 1 ˚C decrease in the operating temperature, the generated power as well as the efficiency increases.
Yanfeng Liu et al. [51]	Num. & Exp.	Active	Cooling system with various parameters	——	The improved cooling system can increase power generation by 7–15 %
Kamal Singh et al. [52]	Exp.	Active	Copper tubes attached behind a conventional photovoltaic panel	The temperature was decreased by 15.23 %	η_{el} increased by 6.08 %.
Linus Idoko et al. [53]	Exp.	Active	Multi-concept cooling techniques (PV + heat sink and water cooling)	The surface temperature was reduced to 20 ˚C	The output increased by 20.96 W, and the efficiency by at least 3 %.

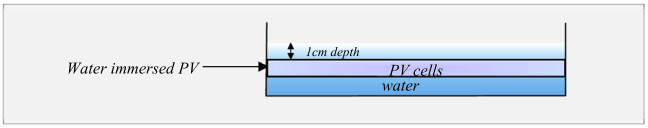


Fig. 6. PV Panel immersed in water [27].

cell under indoor conditions in an array of rectangular microchannels. The data obtained showed that the proposed hybrid system improved the output power by a maximum of 38 %. Rahimi et al. [81] studied the performance of the aforementioned microchannels (see Fig. 12) using water as a single-phase flow. Excellent cooling performance was observed in the experiments, and the cell maximum power was marginally improved above 30 %.

Masoud Rahimi et al. [82] conducted a comparative study on the use of single and multi-header microchannels in the cooling of hybrid photovoltaic panels. The results showed that the heat removed by the multi-header microchannel was approximately 19 % higher than that of the single-header microchannel and an approximately 28 % increase in the electrical power generated by the multi-header microchannel. Another study by, M.R. Salem et al. [83] experimentally examined the performance of photovoltaic cells using straight and helical aluminum microchannel. The results showed that the electrical efficiency increased by an average of 17.7 % to 31.1 % with the straight channel, while the efficiency in the helical channel improved by an average of 20–38.4 % when the flow rate was increased from 0.25 to 1 L/min for both configurations. In the same field, Aly M.A. Soliman et al. [84] numerically studied the performance of a heat spreader and a microchannel to cool solar cells. The results showed that the use of HS with microchannels reduces cell temperature, thermal efficiency, and temperature gradient at the cell surface and increases cell efficiency, electrical power, and net power output.

3.1.9. Jet impingement

Jet impingement water cooling offers an advanced approach to improve photovoltaic system performance. By directing high-velocity water streams directly to the photovoltaic components, this technology efficiently dissipates heat, contributing to effective temperature management and potential efficiency gains. Haitham M.S. Bahaidarah et al. [85] experimentally evaluated the performance of photovoltaic panels using the jet impingement water cooling technique (see Fig. 13). The

Table 3
Summary of most studies on photovoltaic panels with water cooling for PV/T.

Authors	Type of study	Type of cooling	Cooling technique	Thermal results	Electrical Results	Overall results
Yingbo Zhang et al. [54]	Num. & Exp.	Active	porous cooling channel	———	η_{el} improved by 6.08 %.	———
Baloch et al. [55]	Num. & Exp.	Active	Convergent cooling channel	The temperature was significantly reduced from 71.2 to 45.1 °C	The maximum percentage increase in the conversion efficiency was 36.1 %	———
Di Su et al. [56]	Num.	Active	Dual channels for different liquids	———	The air-air PV/T collector results in the highest PV/T temperature and the least electrical efficiency.	The water-water cooling case is the optimum cooling method for improving PV/T generating efficiency
Diego Vittorini et al. [57]	Exp.	Active	Heat recovery system	Thermal recovery for cogeneration purposes is not effective.	The electrical efficiency reached 20 % with an improvement of 33 %	———
Shuang-Ying Wu et al. [58]	Num.	Active	Cooling channel on top of the PV panel	———	The water over the photovoltaic panel resulted in a loss in electrical energy production	The overall energy efficiency was enhanced under all conditions
Ashish Saxena et al. [59]	Exp.	Active	Water cooling system	———	———	The total energy produced increased by about 29 % compared to uncooled PV.
Qi Shi et al. [60]	Num. & Exp.	Passive	PV under the natural water circulation pattern	Annual collector efficiency is 44.2 %	Annual electrical efficiency is 10.2 %	Annual integrated efficiency is 6 %
Afroza Nahar et al. [61]	Num. & Exp.	Active	Parallel plate flow channel	———	———	The numerical and empirical values for the maximum overall efficiency of the PV/T system are found to be 84.4 % and 80 % respectively
Guilherme Zanlorenzi et al. [9]	Exp.	Active	Automatic water-cooling system.	An 11.42 % reduction in average operating temp. which represents a reduction of 8.83 °C	Output power improved by 8.22 %	Average efficiency improved by 33.28 %
Robles-Ocampo et al. [62]	Exp.	Active	PV/T with a transparent collector to use both bifacial PV module	———	The electrical efficiency is estimated at 16.4 %.	The total solar energy utilization efficiency is estimated at 60 %

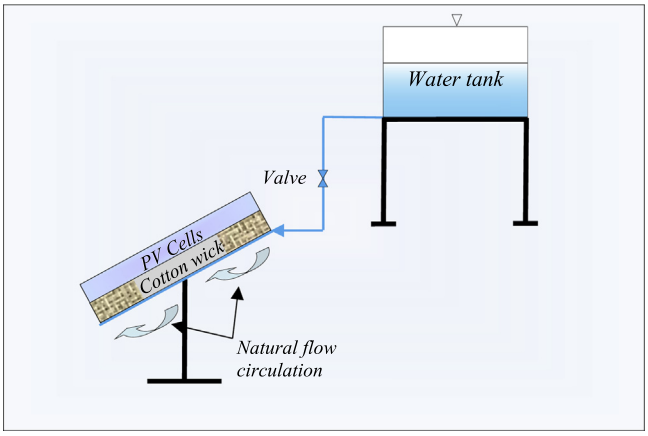


Fig. 7. Schematic diagram of PV cells with an evaporative cooling technique [29].

results showed that the temperature was 69 °C and 47.6 °C for the uncooled system in June and December, respectively. By applying jet cooling, the average cell temperature was reduced to 36.6 °C in June and 31.1 °C. Energy production and conversion efficiency were improved by 51.6 % and 66.6 %, respectively, through the use of jet cooling in June. Similarly, the results showed a performance improvement of 49.6 % in power production and 82.6 % in conversion efficiency in December.

Husam Abdulrasool Hasan et al. [86] designed a photovoltaic solar water collector with a jet impingement system to evaluate its electrical and thermal performance. The results showed that the maximum thermal, photovoltaic, and PVT efficiencies of the PVT collector with jet impingement were 72, 11.35, and 81 %, respectively, at a solar radiation level of 1000 W/m2. Similarly, Javidan et al. [87] experimentally studied the performance of photovoltaic panels combined with a water jet impingement cooling system. The use of PV/JIC reduced the average surface temperature from 63.95 °C to 33.68 °C.

3.2. Air cooling methods

Air cooling is an essential technique for cooling PV systems. This approach effectively uses the thermal properties of air to dissipate heat from the PV components, which contributes to temperature control and system performance. Heat dissipation can be achieved through natural or forced convection mechanisms.

3.2.1. PV with improved design

Optimizing cooling through improved design is a strategic approach for photovoltaic systems. S. Nizetic et al. [88] numerically and experimentally studied a backside convective cooling mechanism. The results showed that the average temperature increase of the PV panel due to the detected effect of flow separation ranged from 5 °C to 9 °C, which means that the estimated degradation of the electrical efficiency of the PV panel could range from 2.5 % to 4.5 % on average. Thus, redesigning the commonly used existing PV panels to break the addressed flow separation can lead to an average increase in electrical efficiency. In the same manner, Abd-Elhady et al. [89] tried to passively cool the PV panel by perforating it with several holes of several diameters (see Fig. 14). The results indicated that the temperature difference increased with an increase in the number of holes. However, there is a critical diameter for the hole size such that the temperature of the photovoltaic panel increases before and after that diameter.

Table 4
Summary of most studies on photovoltaic panels with active spray nozzles water cooling.

Authors	Type of study	Cooling technique	Thermal results	Electrical Results
Azadeh Kordzadeh [68]	Exp.	A thin film of water through a water pumping system	———	The power generated by the array increases significantly
Phayom Warachit [69]	Exp.	Spray nozzles between the front and back side	Front-side cooling is perfectly distributed water over the surface	6.06 % increase in open-circuit voltage and 1.93 % increase in power conversion efficiency compared with no cooling system
Moharram et al. [70]	Exp.	Water-spray cooling system	———	The highest output power is obtained if the cooling is started when the temperature reaches a maximum allowable temperature (MAT) of 45 ˚C.
Raval et al. [26]	Num. & exp.	Thermal energy recovery system by direct contact heat exchange with flowing water from the top of the panel	The overall energy efficiency was increased from about 6 % to 40 %	———
Aldihani et al. [6]	Num.	Cooling the PV with a water temperature of less than 40 ˚C at a speed of 1 mm/*sec*	The panel temperature was reduced by 20 K	The output power improved from 150 W to 171 W at midday.
Y.M.Irwan et al. [71]	Exp.	Water-spray cooling system	Temperature reduced by about 5 - 23 ˚C	The power increased by 9 - 22 %.
Ahmed Elnozahy et al. [72]	Exp.	Cooling and cleaning	Temperature decreased by about 45.5 % and 39 % at the front and back sides, respectively	The efficiency of the cooled and cleaned surface module is 11.7 % versus 9 % for the module without cooling and cleaning (corresponding to percentage improvements of 30 %)
Fabio Schiro et al. [73]	Num. & Exp.	The water spray cooling method	Temperature decreased from 8 to 24 ˚C	———
Jen-Cheng Wang et al. [74]	Theor. & Exp.	Automatic water-cooling system.	———	Energy converted increased by about 17.75 %.
Chia-Yi Mah et al. [75]	Exp.	Introducing a thin film of water over the PV using water-cooling techniques	———	Output power is improved by 15 %, with an optimum cooling rate of 6 L/min
German Osma-Pinto et al. [76]	Exp.	Forced irrigation on the front of the surface	———	The irrigation system can increase up to 10 % of the daily power generation
Piero Bevilacqua et al. [77]	Exp.	Spray water cooling and forced ventilation cooling	———	This system achieved the highest efficiency performance of 14.3 % in August compared to 12.7 % for the reference unit.
Amirhosein Hadipour et al. [78]	Theor. & Exp.	Pulsed spray water cooling system	———	The output power increased by 27.7 % by using only 1/9 of the water required for continuous cooling.
Opeyeolu Timothy Laseinde et al. [79]	Exp.	Automatic water spray cooling system	———	η_{el} of the solar -panel-array increased by 16.65 %.

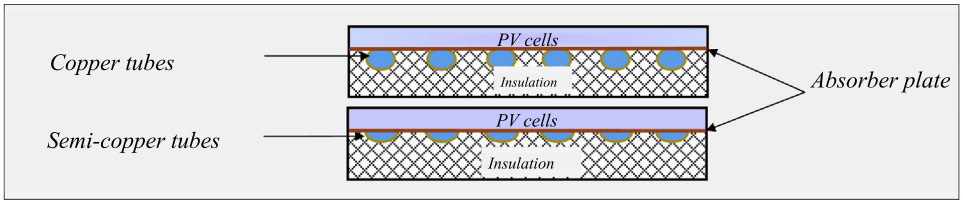


Fig. 8. Schematic representation of solar cell cooling using different serpentine tubes [41].

Similarly, Marinić-Kragić et al. [90] attempted to improve the passive cooling approach to a PV panel by introducing slites. The results showed that the modified module could reduce the PV temperature by up to 3 ˚C.

3.2.2. PV with heat sink

The integration of heat sinks into photovoltaic systems has emerged as an innovative strategy to enhance natural convection. This can be achieved through the incorporation of fins and extended surface designs, often constructed from materials like aluminum with high thermal conductivity. The use of various heat sink types, as depicted in Fig. 15, has been explored to effectively cool photovoltaic cells and optimize their performance (Table 5).

Heat sinks play an important role in achieving consistent passive cooling of PV panels. Factors such as material selection, dimensions, and designs greatly influence their effectiveness. A well-designed heat sink can lower PV panel temperatures by up to 6 ˚C, which can improve electrical performance by over 4 %.

3.2.3. Solar chimney

A solar chimney is a renewable energy technology that uses solar radiation to create an air current through natural convection, which can be used for various purposes, including photovoltaic cooling systems or electricity generation. heng Zou et al. [103] studied the performance of photovoltaic panels installed on a duct that relies on a solar chimney (see Fig. 16) as a passive air-cooling system.

The results showed that, at an ambient temperature of 30 ˚C, the temperature of the PV arrays decreased by 10 ˚C at 500 W/m2 and by 15 ˚C at 800 W/m2.

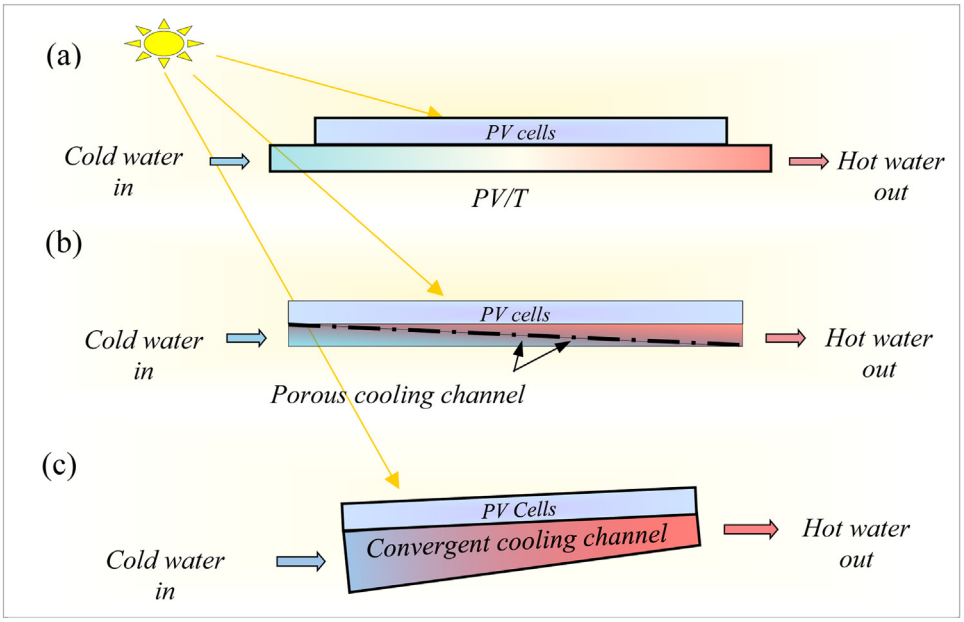


Fig. 9. Schematic diagram of PV with three types of cooling channels: (a) standard cooling channel, (b) porous cooling channel [54], and (c) convergent cooling channel [55].

Table 5
Summary of most studies conducted on photovoltaic panels with different types of heat sinks as a passive cooling method.

Authors	Type of study	Cooling technique	Thermal results	Electrical Results
A.M.Elbreki et al. [91]	Num. & Exp.	Lapping fins combined with a reflective plane	With 18 lapping fins and 27.7 mm fin pitch, the temperature was reduced by 24.57 °C	Electrical efficiency improved by 1.39 % at 1000 W/m2.
Hernandez-Perez et al. [92]	Num. & Exp.	New passive heat sink design	A decrease of up to 10 °C in the most effective dissipator.	An improvement of up to 4 % in electrical efficiency
Ahmad et al. [93]	Num.	A multi-level heat sink (MLFHS)	Temperature reduced by up to 6.13 %	The power output improved by 2.87 %.
Kyu Hyung Do et al. [94]	Exp.	Plate fins for concentrating photovoltaics (CPV).	The optimal spacing of the fins is highly dependent on the temperature difference as well as the angle of inclination	——
Laura Maturi et al. [95]	Num. & Exp.	A heat sink system on the backside of the PV module	Temperature decreased by 5.2 °C	The output power increased by 2.3 %.
Ahmad El Mays et al. [22]	Exp.	Aluminum finned plate at the backside	Temperature decreased by an average of 6 °C.	Electrical efficiency increased by 1.75 % and the output power by 1.86 W.
Filip Grubisic-Cabo et al. [96]	Exp.	Aluminum fins mounted on the backside	The randomly placed finned PV was better than the straight-finned	η_{el} increased by about 2 % compared to the reference PV.
Usman Jamil Rajput et al. [97]	Exp.	Heat sink and water-type PV/T collector	Temperature was reduced by 25 °C using a cylindrical heat sink and 36.5 °C using a single-channel PV/T collector.	——
Ayman Abdel-raheimAmr et al. [98]	Theor. & Exp.	Fins heat sink	The fins result in about a 4–5 °C reduction in cell temperature.	——
Fatih Bayrak et al. [99]	Exp.	Different fin parameters	The temperature was not homogeneously distributed	The highest energy and exergy efficiency was calculated as 11.55 % and 10.91 %, respectively
Fatih Bayrak et al. [100]	Exp.	Various cooling techniques	——	PV/fin produced the highest power generation of 47.88, while PV/PCM -TEM produced the lowest power generation of 44.26 W.
Hernandez-Perez et al. [101]	Exp.	Discontinuous fin heat sink	Temperature reduced by up to 5.5 °C	——
Ellis Johnston et al. [102]	Num. & Exp.	Different fin heights and different angles of inclination	——	The output power increased by 11.3 % to 15.3 % by increasing the heat sink height from 20 mm to 100 mm

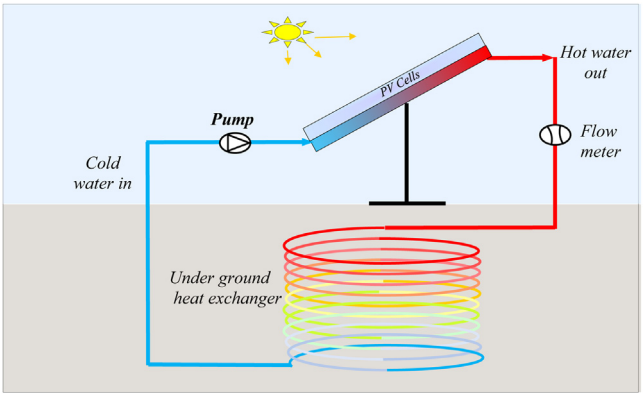


Fig. 10. Schematic of the designed PV panel with earth water heat exchanger cooling system [63].

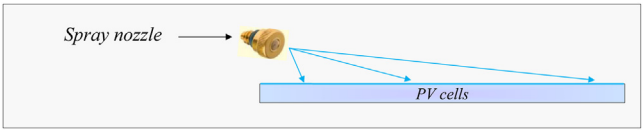


Fig. 11. A PV panel with a specific water nozzle applied to the front side [67].

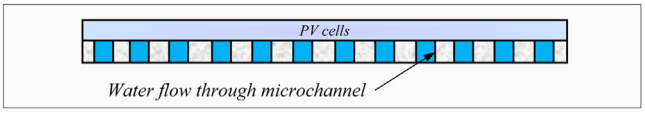


Fig. 12. Scheme of the microchannel water-cooling technique [80,81].

3.2.4. PV/T with air collector

Several types of air collectors have been used to improve the electrical performance of photovoltaic panels and to collect harmful thermal energy from photovoltaic cells. Shuang-Ying Wu et al. [104] numerically studied the effect of air channel location. The results showed that the overall, and thermal efficiency was higher with the cooling channel above the PV panel, while the electrical efficiency was higher with the cooling channel below the PV panel. Fig. 17 shows the three types of air collectors used in previous studies (Table 6).

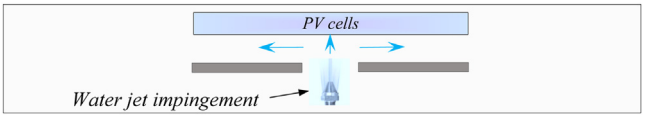


Fig. 13. Schematic of the jet impingement cooling system [85].

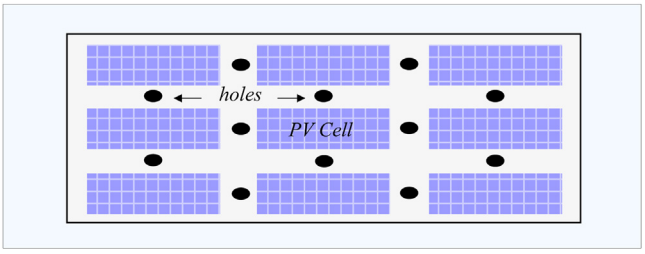


Fig. 14. Schematic of a PV panel with perforated holes [89].

Implementation of specialized cooling techniques for air-type PV thermal panels provides an opportunity to enhance electrical and thermal efficiency. These studies showed different designs of collectors such as honeycomb heat exchangers, single pass collectors with rectangular fins, unglazed transpired collectors (PV/UTC), parallel duct arrays, and other configurations. Temperature regulation can be achieved through air flow rate adjustments - higher flow rates result in lower temperatures and vice versa. A well-designed cooling duct can set a maximum temperature of 45 °C, which can increase the efficiency by more than 4–5 %.

3.2.5. Geothermal air-cooling technique

Geothermal air cooling techniques offer a promising solution for efficient PV cooling systems. By taking advantage of the temperature difference between the ground and the air. Nabil A.S. Elminshawy et al. [114] studied the performance of a buried heat exchanger system (see Fig. 18) for cooling photovoltaic panels under high air temperatures. The results showed that geothermal air cooling resulted in 29.11 %, 23.61 %, and 18.46 % increases in the average daily electricity efficiency when supplying ambient air to (BHE) at temperatures of 35 °C, 40 °C, and 45 °C, respectively.

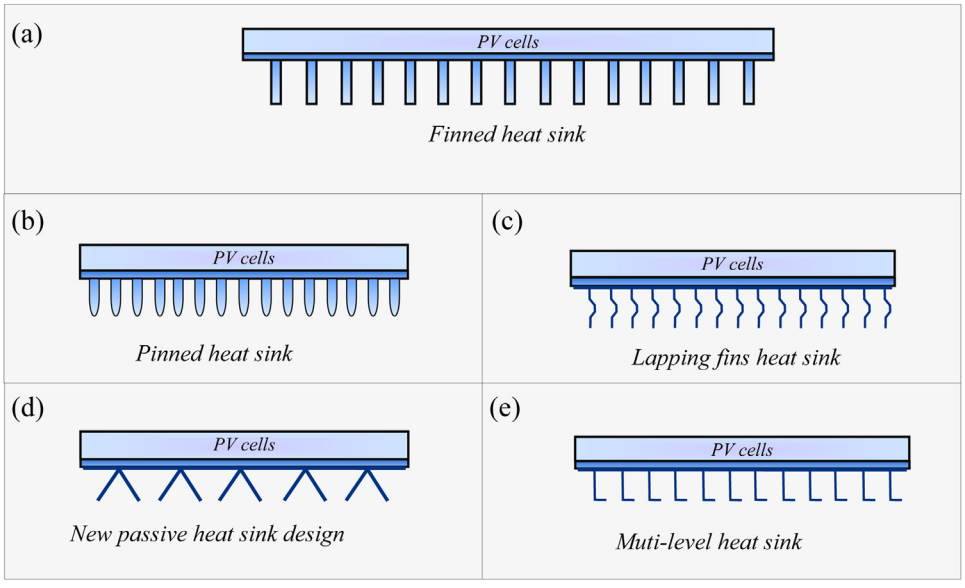


Fig. 15. PV with different types of heat sink: (a) Finned heat sink, (b) pinned heat sink, (c) lapping fins heat sink [91], (d) new passive heat sink [92], and (e) multi-level heat sink [93].

Table 6
Summary of most experimental studies conducted on photovoltaic thermal systems with air collectors.

Authors	Cooling technique	Thermal results	Electrical Results
Hussain et al. [105]	Honeycomb heat exchanger	The thermal efficiency of the system without a honeycomb is 27 % and with a honeycomb is 87 %.	The electrical efficiency of the PV module improved by 0.1 %
Juwel Chandra Mojumder et al. [106]	Air-type single-pass PV/T collector system with several thin rectangular fins	Maximum thermal efficiency was obtained at about 56.19 % for four fins	Maximum electrical efficiency was obtained at about 13.75 %
Shahsavar et al. [107]	(PV/T) systems with two, four, and eight fans operating	Setting glass cover on photovoltaic panels leads to an increase in thermal efficiency and a decrease in the electrical efficiency of the system.	Show that there is an optimum number of fans for achieving maximum electrical efficiency
Teo et al. [108]	A parallel array of ducts with an inlet/outlet manifold that is attached to the back	——	Electrical efficiency increased from 8 to 9 % to between 12 and 14 %
Kaiser et al. [109]	Air channel	——	For a duct velocity $V = 6$ m/s, a 19 % increase in power output was observed
Gholampour et al. [110]	Unglazed Transpired Collector (PV/UTC).	Mounting PV panels on the UTC can result in photovoltaic cooling	——
Alibakhsh Kasaeian et al. [111]	Study the effect of different mass flow rates and channel depths	——	The depth of the air channel did not affect the η_{el} while the mass flow rate had a slight effect on the η_{el} which only changed in the range of 12–12.4 %.
Hasan Saygin et al. [112]	A modified PV/T collector	——	A PV/T system with a gap distance of 5 cm and a mass flow rate of 0.021 kg/s produces the highest η_{el}
Jaafar Hallal et al. [113]	Study the consumption of the cooling system	The maximum allowable temperature is found to be 45 °C by looking at the net output power.	——

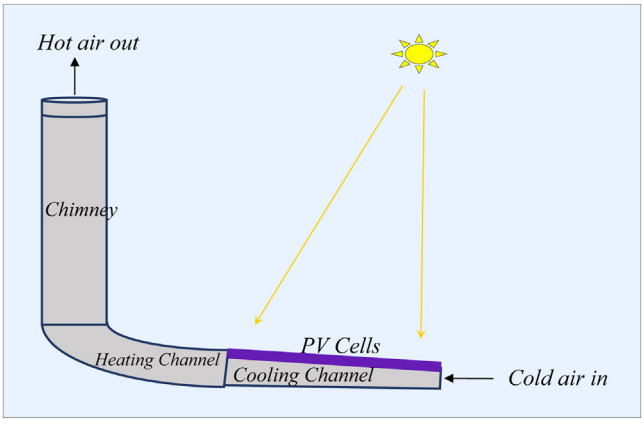


Fig. 16. Geometric model of the duct and solar chimney for cooling PV [103].

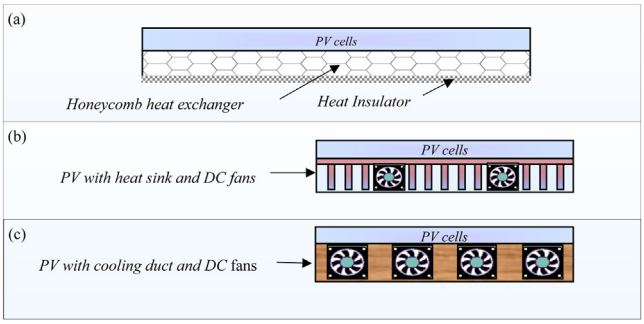


Fig. 17. Schematic for photovoltaic cells with (a) honeycomb heat exchangers [105], (b) finned plate and DC fans [106], and (c) cooling duct and DC fans [107].

3.2.6. Exhaust ventilated air

Cooling photovoltaic systems with exhaust-ventilated air involves utilizing airflow to dissipate heat from panels. A wind-driven ventilator for enhancing photovoltaic cell power generation was investigated by Peyvand Valeh-e-Sheyda et al. [115]. As a result, in addition to normal ventilation by the ventilator, the performance of the photovoltaic cell in terms of energy production was improved by up to 46.54 %. In addition, Shahsavar et al. [116] studied the effects of using exhaust and ventilation air for cooling photovoltaic panels. The results showed that the exhaust and ventilation air in heating ventilating air conditioning systems can be used as the cooling fluid of PV panels and increase their efficiency. On the other hand, the heat rejection of the PV panels could provide some part of the ventilation air-heating load. similarly, Y.B. Assoa et al. [117] numerically and experimentally studied the effect of the air gap ventilation type on solar PV/T hybrid air collector with a metal absorber. The results showed that forced ventilation provided a higher value for heat production, whereas natural ventilation was sufficient

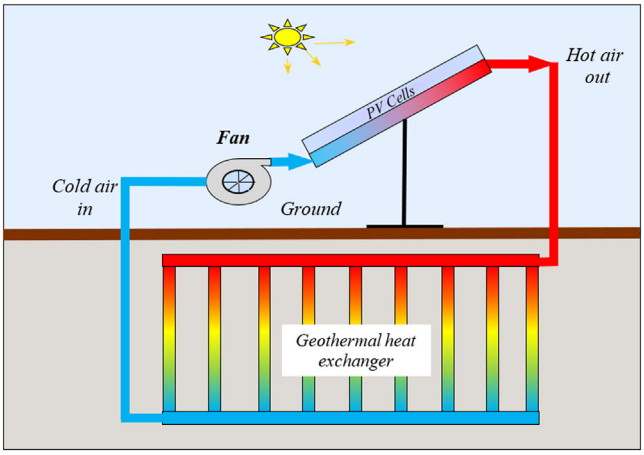


Fig. 18. Buried heat exchanger (BHE) arrangement [114].

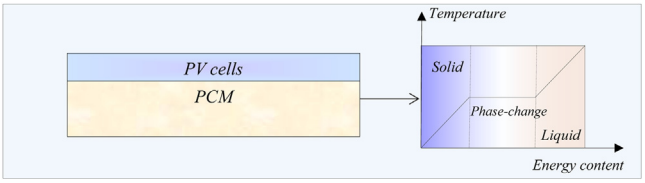


Fig. 19. Schematic for the principle of PV-PCM cooling technique.

to cool the integrated PV modules. Similarly, Rahimi et al. [118] designed and built a conical wind tunnel to cool photovoltaic cells and produce electrical energy via designed turbines. The experimental results showed that the total output power increased by 36 % from both the PV cells and turbine electrical energy production and the numerical results agreed with the experimental results.

3.3. PCM cooling methods

Phase change materials (PCMs) possess a unique property of melting and solidifying at specific temperatures, facilitating the absorption or release of substantial heat energy. This heat absorption is categorized as sensible or latent heat. Sensible heat is absorbed while the material is solid or liquid, while latent heat is absorbed during the transition from solid to liquid states (refer to Fig. 19).

The incorporation of specific PCMs into photovoltaic (PV) panels constitutes a hybrid system capable of passively reducing the surface temperature of PV cells, consequently enhancing their electrical conversion efficiency. In a typical PV-PCM hybrid configuration, the PCM functions similarly to a heat sink, drawing excessive heat from a heated object and mitigating its temperature.

3.3.1. PV with pure PCM

Several types of phase change materials with different melting point ranges under different atmospheric conditions have been used to improve the thermal and electrical performances of photovoltaic cells (Table 7).

The application of PCM can passively lower PV temperatures by more than 10 °C and enhance electrical efficiency by more than 5 %. However, its effectiveness depends on PCM properties and environmental factors. Hasan et al. [128] found reduced cooling capacity during peak hot and cold temperatures due to incomplete phase change. Similarly, Sourav Khanna et al. [132] noted higher efficiency in climates with stable temperatures, lower wind speeds, increased sunlight, and favorable wind directions across PV panels.

3.3.2. PV with composite PCM

The extremely low thermal conductivity of the phase-change material constitutes a thermal insulator rather than absorbing heat from the PV to the ambient environment. Ewald Japs et al. [139] experimentally studied the energy performance and economic yield of a photovoltaic panel with phase change materials (PCMs). The results showed that although a positive increase in the PV output power was observed before noon owing to PCM charging, later in the day, higher operating temperatures of the PV modules were observed because of the thermal insulation effect of the PCM layer attached to the backside. Hence, there was a negative economic yield on most days. In contrast, the PCM with a higher thermal conductivity had significantly lower temperatures after charging and a correspondingly higher yield. Researchers have attempted to solve this problem by improving the thermal conductivity of PCM using nanoparticles. Zigeng Luo et al. [140] tried experimentally and numerically to regulate the temperature of solar panels using form-stable paraffin/expanded graphite composite PCM (see Fig. 20). The results showed that compared with the temperature of the conventional PV panel, the temperature of the PV-PCM panel was kept below 50 °C for 200 min extended by 146 min, with an average resulting power increase of 7.28 % (Table 8).

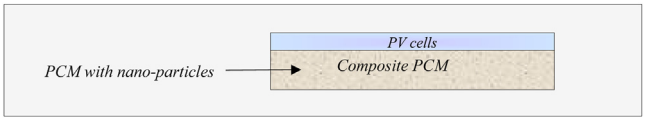


Fig. 20. Cooling of solar cells with composite PCM [140].

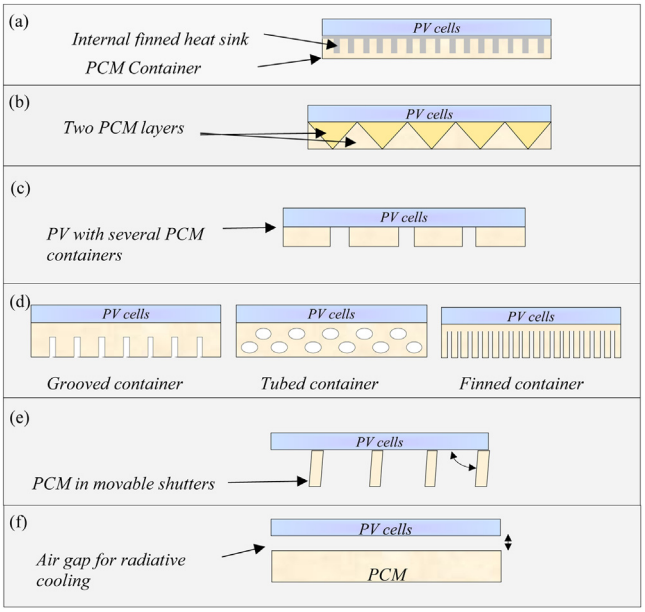


Fig. 21. photovoltaic cells with innovative types of PCM containers: (a) PCM container with internal fins [148], (b) triangular metal cells for different PCMs [149], (c) several PCM containers [150], (d) grooved, tubed, and finned container [151], (e) PCM in movable shutter [152], and (f) PV with indirect contact with the PCM container [153].

Enhancing the thermal conductivity of PCM through the introduction of nanoparticles, aluminum fins, copper foam matrix, or other materials has the potential to significantly improve the cooling effectiveness of the PV-PCM system, ultimately leading to a potential overall efficiency increase by more than 6 %.

3.3.3. PV with PCM and heat sink

Other researchers have used PCM with innovative enclosures to enhance the thermal behavior of PCM, such as the use of internal fins [148], two PCM enclosures [149], several PCM enclosures [150], grooved, tubed, fin enclosures [151], movable shutters [152], and many other designs, as summarized below (see Fig. 21) (Table 9).

These studies have shown several PCM enclosure designs that have the potential to enhance heat transfer between the PCM and the ambient temperature, resulting in more efficient cooling effects. A well-designed PCM enclosure has the potential to increase the electrical efficiency of a PV/PCM passive cooling system by more than 4 %.

3.3.4. PV with PCM and water cooling

The use of phase change material with water-cooling techniques (see Fig. 22) has resulted in an efficient cooling technique that can maintain the PCM temperature in the melting range, thus maintaining the PV temperature at a low and stable temperature. This technique can absorb heat from photovoltaic cells, maintaining a good heat distribution in addition to making use of the hot water produced (Table 10).

The presented studies explore the integration of phase change materials and different cooling techniques to improve photovoltaic panel performance. Through active and passive cooling strategies, the researchers

Table 7
Summary of most studies conducted on photovoltaic panels with pure PCM as a passive cooling technique.

Authors	Type of study	Cooling technique	Types of PCMs and their melting point	Thermal results	Electrical Results
Hassan et al. [119]	Exp.	Phase change material	Capric–lauric acid, (20.78 °C), capric–palmitic acid, (22.33 °C), Pure salt hydrate (29.17 °C), and Commercial blend PCM (22.97 °C)	The temperature was lowered to 18 °C for 30 min while it was lowered to 10 °C for 5 h using salt hydrate PCM at 1000 W/m2 insolation.	———
Smith et al. [120]	Num.	Study the performance of varying PCM melting temperatures from 0 °C to 50 °C in different regions around the world	———	———	The annual PV energy output increases by over 6 % in Mexico and eastern Africa, and over 5 % in many locations. In Europe, the energy output enhancement varies between 2 % and nearly 5 %.
Jungwoo Park et al. [121]	Exp.	Vertical PV module with the application of PCM	Paraffinic hydrocarbon PCM (24.85 °C)	———	Electric power during the winter season increased by 1.0–1.5 %
Hagar Elarga et al. [122]	Num.	PV-PCM system at double skin facades	RT42-Organic material (38 °C), and RT55-Organic material (51 °C)	———	The electrical efficiency improved in the range of 5–8 %
Rok Stropnik et al. [123]	Num. & exp.	PV with PCM	RT28HC (28 °C)	The maximum temperature difference was 35.6 °C higher than that of the PV/PCM in one day	———
Anna Machniewicz et al. [124]	Num.	Studied the effect of transition temperature on the efficiency of PV/PCM	RT10HC (9 °C), RT18HC (17 °C), RT25HC (22 °C), and RT35HC (34 °C)	PCM with a transition temperature of about 20 °C can effectively increase the η_{el}	———
Ahmad Hasan et al. [125]	Exp.	Phase change material in hot climatic conditions	RT42 (41 °C)	The indoor cooling effect improved by 9.5 % at peak and 7 % on average during the day.	PV output increased by 7.2 % at peak and 5 % on average
Laura Aelenei et al. [126]	Num. & Exp.	Building-integrated photovoltaic panel with PCM	PCM, Alba®balance type with freezing point 18 °C	The maximum thermal efficiency can reach 12 %.	The maximum electrical efficiency can reach 10 %.
Kibria et al. [127]	Num.	PV/PCM integrated with buildings.	———	The PCM is shown to be an effective way to limit the temperature rise in PV devices	PCM enhanced the efficiency of PV modules by up to 5 %.
Hasan et al. [128]	Num. & Exp.	PCM in different climate conditions.	RT42 (38–43 °C)	PCM produced less cooling in the cold peak and the hot peak due to incomplete melting and solidification, respectively.	The PV-PCM system boosted annual PV energy production by 5.9 % in hot weather conditions.
Mousavi Baygi et al. [129]	Exp.	PV with PCM	Polyethylene glycol 1000 (PEG 1000) with a melting point of 38 −40 °C	———	The electrical efficiency of the system improved by up to 8 %.
Sourav Khanna et al. [130]	Num.	PV/PCM under variable operating conditions.	RT25HC (26.6 °C)	Adding PCM decreased temperature by 19 °C	PCM improved the electrical efficiency from 17.1 % to 19 %. (corresponding to percentage improvements of 11.11 %)
Sandro Nižetić et al. [131]	Num.	Pork fat as a potential new PCM	Pork fat (36 - 45 °C)	———	There is no significant difference in the performance benefit as compared to conventional organic PCM.
Sourav Khanna et al. [132]	Num. & Exp.	Studied the climatic effects on PV/PCM	RT 25 HC (25.6 °C)	The PCM is more effective in climates with less ambient temperature variation, lower wind speed, and higher solar irradiance, and in places where the wind flow is across the PV	The efficiency of PV/PCM decreased from 9.7 % to 6.6 % due to significant climatic changes.
Sourav Khanna et al. [133]	Num.	PV/PCM system	RT 25 HC (25.6 °C)	———	With increasing wind azimuth angle from 0° to 90°, the optimum depth of the PCM enclosure increases from 3.9 cm to 5.3 cm
Müslüm Arıcı et al. [134]	Num.	PCM	Organic PCMs with diffirent melting points	PCM reduces the operating temperature of the photovoltaic panel by up to 10.26 °C. They also found the optimum PCM melting point and thickness at two different locations throughout the year	The efficiency increased by up to 3.73 %,
Razali Thaib et al. [135]	Exp.	Beeswax as PCM	paraffin wax (46.7 °C), and beeswax (51 °C)	———	Beeswax has the required application potential with efficiency ranging from 7 % to 7.8 %.
Jiaxin Zhao et al. [136]	Num.	PV/PCM system	RT35HC (36 °C)	Temperature could be reduced by 24.9 °C	The output power could be increased by up to 11.2 %.
Karthikeyan Velmurugan et al. [137]	Exp.	Eutectic PCM	Lauryl Alcohol (25.91 °C), Cetyl Alcohol (51.5 °C), and an eutectic PCM mixture of 75 % Lauryl - 25 % Cetyl (21.75 °C).	Eutectic PCM can reduce the photovoltaic temperature by 10.3 °C	The electrical efficiency improved by 10.09 %.
Mohammad Rezvanpour et al. [138]	Num. & Exp.	CaCl2 6 H2O as PCM	Calcium chloride hexahydrate ($CaCl_2 \cdot 6 H_2O$) with a melting point of 29.6 °C	The maximum temperature dropped was 26.3 °C (38 % decrease)	The power increased by 1.16 W (24.68 % of the increase).

investigated the effect of factors such as PCM melting point, nanofluid utilization, water flow rates, and cooling system design on both thermal and electrical efficiencies. Notable results include the ability of the water-PCM to stabilize the PV temperature, resulting in gains in electrical power by more than 48 %.

4. Other cooling methods

Ongoing research in the field of renewable energy, especially in the cooling of photovoltaic panels, has developed many new techniques that have the potential to lower the photovoltaic temperature and improve

Table 8
Summary of most studies conducted on photovoltaic panels with composite PCM as a passive cooling technique.

Authors	Type of study	Cooling technique	Thermal results	Electrical Results
Leila Siahkamari et al. [141]	Exp.	Sheep fat + CuO nanoparticles	——	The maximum power generated increased by about 24.6 % to 26.2 % compared to the no-cooling system and 5.3 % to 12 % compared to paraffin wax.
Peter Atkin et al. [142]	Exp.	PCM-infused graphite and aluminum fins	——	The overall efficiency increased by about 12.97 % while the least improvement was found by PCM or fins alone.
Sharma et al. [143]	Exp.	Nano-PCM with micro-fins	The average temperature decreased by 10.7 °C using PCM/micro-fins and 12.5 °C using n-PCM/micro-fins compared to using micro-fins only. Similarly, the temperature decreased using PCM and n-PCM by 9.6 °C and 11.2 °C, respectively, compared to the natural convective condition.	——
Farouk Hachem et al. [144]	Exp.	Pure and combined phase change materials	——	The η_{el} increased by an average of 3 % when pure PCM was used and by an average of 5.8 % when using combined PCM.
Al-Waeli et al. [145]	Exp.	PV/T with nanofluid and nano-PCM	——	Nano-PCM and nanofluid improved the electrical efficiency from 8.07 % to 13.32 %. (corresponding to percentage improvements of 65 %)
Abdulmunem et al. [146]	Exp.	PCM with a copper foam matrix and multi-walled carbon nanotubes	——	Using PCM only increased the efficiency by 1.97 %, while the efficiency increased by 5.68 % using PCM/copper foam matrix and multi-walled carbon nanotubes
Al-Waeli et al. [147]	Exp.	Nano-enhanced-paraffin and nanofluids	——	The highest efficiency of the photovoltaic module was 13.7 %, while the efficiency of the conventional module was 7.1 %. (corresponding to percentage improvements of 92 %)

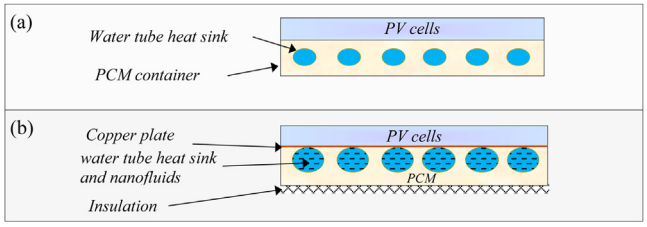


Fig. 22. Photovoltaic cells with PCM and (a) water tube heat sink [164], (b) water tube heat sink and nanofluids [165].

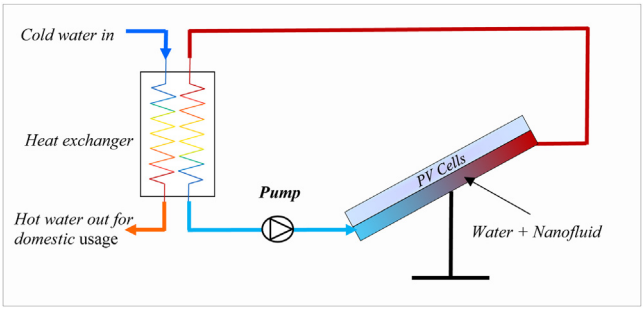


Fig. 23. Schematic of nanofluid-cooled PVT system [178].

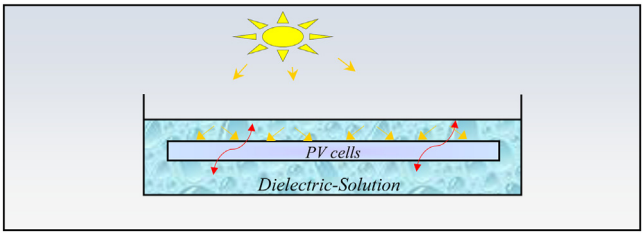


Fig. 24. Schematic diagram for cooling PV cells in a dielectric solution [196].

its performance. such as using nanofluids as coolants, thermoelectric cooling, liquid immersion, radiative cooling, heat pumps, heat pipes, and many other cooling techniques, are summarized below.

4.1. Nanofluid as coolants

Nanofluids play an essential role in improving the heat-transfer rate of some fluids. The use of this material with a small concentration of nanoparticles in water-cooled photovoltaics or even PCM-cooled photovoltaics enhances the efficiency of the system. However, to take advantage of the waste heat, an external heat exchanger must be added to extract heat from the nanofluid, as well as to re-cool it again to complete the cycle (see Fig. 23) (Table 11).

These studies investigate the integration of nanofluids of different concentrations into various cooling technologies to enhance the efficiency of the photovoltaic system, with a focus on active cooling methods. Nanofluids such as silver nanoparticles, aluminum oxide, titanium dioxide, graphene, etc. are combined with water-based coolants. The results show elevated thermal and electrical efficiency, lower cell temperature, and improved overall performance.

4.2. Liquid immersion cooling

Liquid immersion cooling is a technology used to efficiently remove heat from photovoltaic panels by submerging them in a dielectric solution that possesses specific properties such as high thermal conductivity, low viscosity, long-lasting chemical and physical stability, low optical absorbance, and good optical stability, as shown in Fig. 24. Li Zhu et al. [196] proposed a liquid immersion cooling method to effectively re-

Table 9
Summary of most studies conducted on photovoltaic panels with innovative PCM enclosures as passive cooling techniques.

Authors	Type of study	cooling techniques added to PV	Thermal results	Electrical Results
Tao Ma et al. [148]	Num.	PCM with a finned heat sink chamber	The highest performance was obtained by PCM called RT35HC compared to RT35 and RT45, which succeeded in maintaining the temperature below 45 °C for 318 mins.	The electricity production increased by 6.35 % and 3.52 % in 3 and 6 h, respectively.
Ming Jun Huang [149]	Num.	Two sets of different PCMs are connected to the PV module in a triangular shape	This system could mitigate the volume expansion and prolong the thermal regulation period which could keep the PV at the operating temperature closer to its characteristic value of 25 °C	This system could improve solar-to-electrical conversion efficiency
Sandro Ni˘zeti´et al. [150]	Exp.	PCM is filled with several small aluminum enclosures	——	PV performance improved by 10.7 %
Wongwuttanasatian et al. [151]	Exp.	PCM filled into various housings, grooved, tubed, and finned	PV-PCM with a finned enclosure lowered the temperature by 6.1 °C	PV-PCM with a finned enclosure improved the electrical efficiency by 5.3 %
Adeel Waqas et al. [152]	Num.	PCM with movable shutters	This technique could reduce the temperature from 64 °C to 42 °C	Efficiency can be improved by up to 9 % during the summer.
Karthikeyan et al. [153]	Num.	PCM was placed under the PV with indirect contact with it to avoid the effect of the low thermal conductivity of the PCM	The temperature of the PV module was 47.81 °C	The efficiency of the PV module was 14.75 %
Huang et al. [154]	Exp.	Study the effect of convection in a finned PV/PCM system	The addition of internal fins improves the temperature control of the PV in a PV/PCM system.	——
Hasan et al. [155]	Num. & Exp.	Two types of PCM: salt hydrates, CaCl2–6H2O, and a eutectic mixture of fatty acids, capric acid-palmitic acid	The best results were found by CaCl2–6H2O which lowered the photovoltaic temperature by 3–4 °C	CaCl2–6H2O resulted in a 3 % greater energy saving compared to capric-palmitic acid in Vehari., Pakistan.
Lippong Tan et al. [156]	Exp.	Enhanced PCM and metal fins	The temperature was reduced by 15 °C	The electrical efficiency improved by 5.39 %
Rajvikram et al. [157]	Exp.	Different cooling methods	The maximum output power of about 1.48 W was produced using the PCM and finned heat	The maximum efficiency using PCM and finned heat is 17.03 %
Rajvikram et al. [158]	Exp.	PV-PCM with aluminum plate	——	The conversion efficiency improved by 24.4 %
Preeti Singh et al. [159]	Exp.	PCM with conductivity-enhancing containers	Temperature can be reduced from 77 °C to 54 °C	Electric generation can be increased by 6.2 % in January and 8.3 % in June
Rajan Kumar et al. [160]	Exp.	External fins attached to an aluminum enclosure for PCM	——	Electrical efficiency improved by 5.16 % using PCM, and by 6.59 % using PCM and external fins
Mohamed Sharaf et al. [161]	Exp.	PCM with aluminum metallic foam AMF	The temperature in December, January, and February decreased by 4 %, 7.4 %, and 13.2 %, respectively	Power in December, January, and February increased by 1.85 %, 3.38 %, and 4.14 %, respectively
Pascal Henry Biwole et al. [162]	Num.	Impure PCM with aluminum fins	Adding PCM can keep the panel operating temperature under 40 °C for 80 min under a constant solar irradiance of 1000 W/m2.	——
Mohammad Amin Vaziri Rad et al. [163]	Exp.	PCM and aluminum shavings porous media	The porous media of aluminum shavings reduced the PCM melting time by about 19–25 %, which reduced the temperature by 24 °C	Electrical power increased by 17.7 %, and electric efficiency improved by 2.5 %.

move heat from densely packed solar cells in highly concentrated systems. The study showed that the module temperature can be cooled to less than 45 °C.

Xinyue Han et al. [197] conducted a numerical and experimental study to demonstrate that immersion cooling using specific liquids such as deionized water, isopropyl alcohol, ethyl acetate, and dimethyl silicon can increase the electrical performance of composite photovoltaic silicon cells while maintaining a low and uniform cell temperature. The results showed that a 1.5 mm thick liquid layer over the cell surface increased the efficiency from 8.5 % to 15.2 %. Similarly, Yu.A. Abrahamyan et al. [198] conducted a study on improving the efficiency of solar cells, by immersing them in an isotropic dielectric fluid. They found that the presence of a dielectric film can increase the efficiency of solar cells by 40–60 % compared to the reference value. Moreover, Subarna Maiti et al. [199] studied the performance of cooling the concentrated photovoltaic panel by using a suitable liquid for the heat exchanger, using a square parabolic-type reflector. The results showed that a more than two-fold increase in output power was realized on a clear sunny day employing a 0.13 m^2 silicon solar module.

4.3. Radiative cooling (optical filtration)

Radiative cooling is one of the most promising passive cooling technologies that has the potential to improve the performance of photovoltaic cells without using any external energy. This technology is based on changing the spectrum of a photovoltaic module to selective spectral thermal radiation with an infrared spectrum range of 8–14 μm that releases heat from terrestrial objects to the outside without absorption from the surrounding environment. Guanheng Fan et al. [200] designed a photonic cooler (see Fig. 25) to cool the solar cells of a space solar power plant by selectively reflecting solar radiation and enhancing the radiative cooling to outer space. This technique can effectively reduce

Table 10
Summary of most studies conducted on photovoltaic panels with PCM and water-cooling techniques.

Authors	Type of study	Type of cooling	Cooling techniques added to PV	Thermal results	Electrical Results
Di et al. [164]	Num.	Active	PV/T with PCM	With the PCM melting point at 40 °C, the PV/T collector obtained the maximum overall energy output	With the PCM melting point at 30 °C, the PV/T obtained the best electrical performance.
Mohammad Sardarabadi et al. [165]	Exp.	Active	Using ZnO/water-nanofluid and PCM	——	The average electrical output increased by more than 13 % compared to the conventional PV module
Malvi et al. [166]	Num.	Active	PV/PCM	With an average water temperature rise of 20 °C, the PV output can be increased by typically 9 %	Any increase in PV performance has an associated decrease in solar thermal performance.
Wang et al. [167]	Exp.	Active	BIPV/T system consists of PCM and metal wires that fill the space between the finned heat pipes and the thermal insulation	The average daily thermal and overall efficiency was 61.1 %, and 68.9 %, respectively, at maximum simulated solar radiation of 900 W/m2 and a water flow rate of 600 L/h. the maximum water tank temperature can reach about 47.23 °C under the same conditions	The average daily electrical efficiency was 7.8 % under the same conditions.
Browne et al. [168]	Exp.	Active	PV/T-PCM	The temperature reached by the water was about 5.5 °C higher when compared to the PV/T system without PCM.	——
Browne et al. [169]	Exp.	Active	PCM and water cooling through a network of tubes placed inside a PCM container	The PV/T-PCM system can increase its heat storage compared to the PV/T system under simulated conditions.	——
Ho et al. [170]	Num.	Active	BIPV combined with the MEPCM	The highest average surface temperature of the PV/MEPCM cell was 34.1 °C	The total electric generating capacity for a day was 231.439 kJ/m, which represented an increase of 2.013 %
Ewa Klugmann-Radziemska et al. [171]	Exp.	Active	PCM	PCM could keep the PV panel at a lower temperature for more than 5 h.	——
Sajan Preet et al. [172]	Exp.	Active	PV/T-PCM	——	The maximum increase in electrical efficiency was 10.66 with water-based PV/T and 12.6 with water-based PV/T-PCM
Salem et al. [173]	Exp.	Active	Al2O3/PCM mixture and/or water cooling-techniques	——	Al2O3 (φ = 1 %) / PCM mixture (λPCM = 25 %) + 75 % water achieved the highest PV performance.
Nasef et al. [174]	Num.	Active	CPV-PCM	The proposed system achieved a 60 % reduction in the average temperature and lowered the PV temperature by 4 °C	The use of nanofluids increased the efficiency of CPV systems by 2.7 %
Nasrin Abdollahi et al. [175]	Exp.	Passive	PV/nano-PCM with water cooling	——	Composite oil as PCM enhanced the output power by 29.24 % at 690 W/m2. While adding Boehmite nano-powder to PCM enhanced the output power by 48.23 % at the same radiation intensity.
Hongtao Xu et al. [176]	Exp.	Active	PCM/water	Results showed that by setting the temperature at 45 °C the total efficiency can reach 91 % while by setting the temperature at 50 °C the overall efficiency can reach 85 %.	——
Sudhakar et al. [177]	Exp.	Active	OM 35 PCM with a water-cooling system	This system could lower the cell temperature by 5.4 °C	The output power and the electrical efficiency could be improved by 13.54 % and 12.4 %, respectively.

the temperature of solar cells by 30 °C and increase their efficiency by 1.4 %.

Rosa-Clot et al. [201] proposed a novel solution to extract infrared radiation while keeping the visible part almost unaffected by a superimposing water layer on the PV panel. This innovative setup, known as Thermal Electric Solar Panel Integration (TESPI), has demonstrated the potential to enhance PV efficiency and heat production. In a similar vein, Ahmed S. Abdelrazik et al. [202] investigated the effects of optical filtration and nano-phase change materials (nano-PCMs) on the performance of photovoltaic solar collectors. The results indicated that nano-PCM usage only resulted in a minor reduction of approximately 6.7 % in overall efficiency, while nanofluid application for optical filtration led to an improvement in efficiency ranging from 6 to 12 %. Moreover, Zhang et al. [203] explored a blended Ag nanofluid optimized for optical properties to regulate the temperature of photovoltaic/thermal (PV/T) systems. The experiments revealed that the electrical efficiency of the PV/T system was 11.85 % for the 20 nm nanofluid and 11.0 % for the 20/40 nm nanofluid, while the efficiency of the gallium arsenide (GaAs)

Table 11
Summary of most studies conducted on photovoltaic panels with water-nanofluids cooling techniques.

Authors	Study type	Type of cooling	Cooling method	Types of nanofluids	Thermal results	Electrical results
Lari et al. [178]	Num.	Active	PV/T with water nanofluid	Silver nanoparticles and potassium oleate surfactant	Thermal output increased by 13 %.	Electrical output increased by 8.5 %
Mohammad Sardarabadi et al. [179]	Num. & Exp.	Active	Nanoparticles are dispersed in deionized water as a base liquid	Al2O3, TiO2, & ZnO	———	TiO2/water and ZnO/water offer better performance in terms of electrical efficiency
Nooshin Karami et al. [180]	Exp.	Active	Based on helical and straight cooling channels	Boehmite nanofluid (AlOOH-xH2O)	Temperature was reduced by 39.7 % for straight and 53.76 % for helical channel	Electrical efficiency improved by 37.67 % for helical, and 20.57 % for straight channel
Jee Joe Michael et al. [181]	Exp.	Active	New design of PV/T collector with nanofluids	copper oxide water (CuO/H2O)	Significantly improve thermal efficiency	Can improve electrical efficiency
Khanjari et al. [182]	Num.	Active	PV/water-nanofluids	Al2O3-water, Ag-water	Alumina-water improved heat transfer by 8–10 %, Ag-water by 28–45 %	Overall exergy efficiency is not that satisfactory
Abballah et al. [183]	Exp.	Active	PV/T with nanofluid	Al2O3 nanofluid	———	The average and maximum electrical efficiency were improved by 13.58 % and 16.4 %, respectively, at an optimum flow rate of 1.2 L/min.
Ali Radwan et al. [184]	Theor. & Num.	Active	Microchannel heat sink with nanofluids	(Al2O3) & (SiC)	The solar cell temperature was reduced to 38 °C	Electrical efficiency improved by up to 19 %.
Al-Waeli et al. [185]	Exp.	Active	PV/water-nanofluids	Al2O3, CuO, and SiC	Silicon carbide has the best stability and highest thermal conductivity.	———
Al-Waeli et al. [186]	Exp.	Active	PV/ water-nanofluids	SiC nanofluid	The thermal efficiency increased by up to 100.19 % compared to the use of water for cooling.	The use of 3 wt % SiC nanofluid increased the electrical efficiency by up to 24.1 % compared to the PV system alone
Al-Waeli et al. [187]	Exp.	Active	PV/T-nano PCM-nanofluid system	Nano-SiC particles	Thermal efficiency improved to 72 %.	Power increased from 61.1 W to 120.7 W (97 % of the increase), and electrical efficiency from 7.1 % to 13.7 %. (92 % of the increase)
Al-Waeli et al. [188]	Exp.	Active	PV/T system containing SiC nanofluids as coolant	SiC nanofluids	Results showed a decrease in thermal conductivity within days. Therefore, it is better to remix the water-nanofluid solution every three months to avoid nanoparticle precipitation.	———
Ebaid et al. [189]	Exp.	Active	PV with (TiO2) nanofluid in a water-polyethylene glycol mixture and (Al2O3) nanofluid in a water cetyltrimethylammonium bromide mixture	TiO2 & Al2O3 nanofluid	Both types caused a higher decrease in the average temperature of the photovoltaic cell. In addition, the Al2O3 nanofluid showed better performance than the TiO2 nanofluid	———
Nasrin et al. [190]	Num. & Exp.	Active	PV/T integrated with the water-nanofluid/MWCNT system	MWCNT nanofluid	———	Photovoltaic performance improved by 9.2 %
Moh et al. [191]	Exp.	Active	PV with nanofluid and microchannel	Graphene nanoparticles (GNP) nanofluid	At 1.11 L/min, the operating temperature was lower than 45 °C	The highest effective efficiency is 42 % compared to water-cooled and 54 % compared to un-cooled.
Xinyue Han et al. [192]	Num.	Active	CPV/T with a nanofluid-based spectral beam splitter and heat pipe cooling	Ag/CoSO4-propylene glycol (PG) nanofluid	———	The average overall efficiency was 53.66 %.
Saber Abdo et al. [193]	Exp.	Passive	PV/ hydrogel beads with an alumina water-based nanofluid	Alumina nanofluid	Alumina-water-based nanofluid could significantly reduce the temperature by 17.9 °C and 16.3 °C at radiation intensities of 800 and 1000 W/m2, respectively	The optimized concentration could enhance the electricity efficiency of the panels by approximately 10.1 %
Jinxin Feng et al. [194]	Exp.	Active	PV with nano emulsions	OP35E-nanoemulsions	The temperature was reduced by 5.3 % compared to water cooling, and the pumping power was reduced to 50 % to get the same performance as that of water cooling.	———
Amin Shahsavar et al. [195]	Exp.	Active	PV with water magnetite nanofluid through a finned and un-finned thermal collector	FeCL3·6H2O	———	Efficiency improved by 12.06 %, 10.87 %, and 8.40 % for PVT with 8 fins and serpentine tubes, PVT with 4 fins and serpentine tubes, and PVT with serpentine tube systems, respectively.

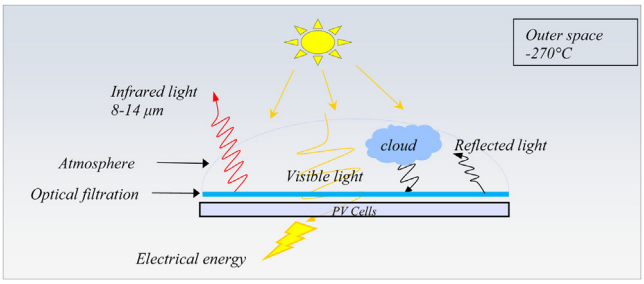


Fig. 25. Schematic diagram for cooling of solar cells with photonic cooler [200].

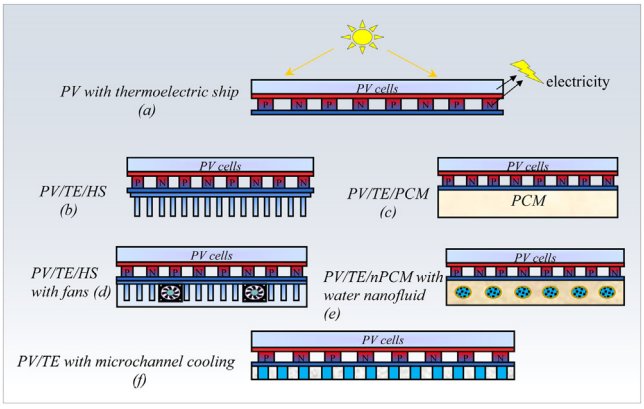


Fig. 26. Photovoltaic cells with thermoelectric ship alone (a) [206] and with different cooling techniques: (b) with finned heat sink [207], (c) with PCM [208], (d) with finned heat sink and DC fans [23], (e) with water-nanofluid and nano-PCM [209], and (f) with microchannel water cooling.

cell was 9.30 % for the 20 nm nanofluid and 9.03 % for the 20/40 nm nanofluid. Song et al. [204] proposed a photovoltaic-thermoelectric radiative cooling system (PVRC-TE) that has been demonstrated to significantly reduce the temperature of PV modules and improve their electrical performance. Additionally, Li et al. [205] conducted a study on the passive cooling performance of photovoltaic panels, which found that selective spectral, passive radiative, and combined spectral and passive radiative cooling could increase efficiency by 0.98 %, 2.40 %, and 4.55 %, respectively.

4.4. Thermoelectric cooling

This technique converts harmful heat into electrical energy via the Peltier effect. More precisely, the thermoelectric generator uses the temperature difference across the front and back to generate electricity, and the greater the temperature difference, the greater the power generation. As a result, this approach has been utilized to cool photovoltaic cells passively, either independently or in conjunction with various cooling techniques (refer to Fig. 26) (Table 12).

These studies explore the integration of passive thermoelectric systems and diverse cooling techniques to enhance photovoltaic system performance. The findings highlight significant improvements in temperature control, efficiency, and power output.

4.5. Heat pipe

This technique is based on the absorption and release of energy from a specific fluid within a heat pipe system (Fig. 27). When heat is applied to one end of the tube, the liquid inside it immediately evaporates and travels to the other cold side of the tube, allowing for rapid condensation of the vapor. Thus, this technique passively releases the absorbed

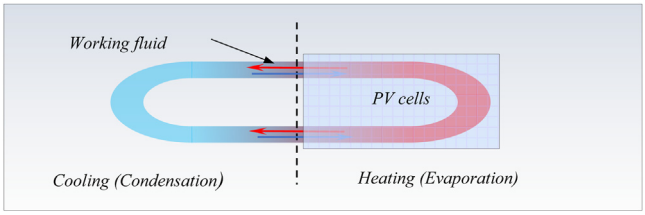


Fig. 27. PV panel with a single-turn pulsating heat pipe [217].

heat during evaporation, which is suitable for cooling photovoltaic cells (Table 13).

These studies explore the utilization of heat pipes as a passive cooling method to enhance photovoltaic system's performance, leading to improvements in both thermal and electrical aspects. Notable examples include a single-turn pulsating heat pipe (PHP) showing an 18 % increase in electrical power, and PV/T systems with heat pipes achieving up to 45.14 % thermal efficiency improvement. These studies confirm the potential of using heat pipes as cooling techniques to optimize both the thermal and electrical aspects of PV systems.

4.6. Heat pump

The integration of the photovoltaic and heat pump systems improved the efficiency of both systems, and the photovoltaic cells were cooled to increase the energy conversion efficiency while providing heat energy to the heat pump evaporator. Xudong Zhao et al. [226] theoretically studied the performance of using the PV module as an evaporator for heat pump operation. The results showed an achievement of 55 % in thermal efficiency and 19 % in electrical efficiency (Table 14).

These studies demonstrate the integration of heat pumps with PV panels to enhance the electrical efficiency of PV systems. By combining PV system with heat pump, high generation efficiency and peak power are achieved.

4.7. Other uncategorized cooling methods

Other scientists have attempted to cool photovoltaic cells using different techniques. Some of these techniques are summarized in Table 15.

This group of studies delves into innovative active cooling methods aimed at improving photovoltaic systems performance. Researchers investigate various approaches, including ferrofluids with magnetic fields, high-frequency ultrasound, flipping technique, and others. Noteworthy results include significant temperature reductions, increased power output, and overall efficiency improvements.

5. Discussion

The literature offers various effective ways to cool PV panels efficiently, which could significantly improve their electrical efficiency. This review's main goal is to identify and highlight the most promising techniques that deserve further research. Cooling solar panels with water shows potential for boosting their efficiency. Methods like water spraying, immersion, circulating liquids through tubes or microchannels, water jet impingements, and evaporative cooling demonstrate efficiency gains of 13 % to 66 % compared to other approaches (Fig. 28). Although there are costs associated with setting up water systems, including expenses for pumps, piping, and equipment, their effectiveness in dusty or sandy areas to support high efficiency is undoubtedly compelling. Moreover, most of these techniques offer the advantage of reusing the extracted heat for domestic or other applications, making them more viable and efficient.

The literature on air cooling techniques reveals that different methods of cooling PV systems are being explored. These methods include different designs of heatsinks, fans, ducts, and finned plates, which have

Table 12
Summary of most studies conducted on photovoltaic panels with thermoelectric cooling techniques.

Authors	Study type	Type of cooling	Cooling method	Thermal results	Electrical results
Fallah Kohan et al. [206]	Num.	Passive	PV/TE	———	More power can be generated. However, the increase is marginal due to the insufficient temperature difference for the TEG device
Wei Pang et al. [207]	Exp.	Passive	PV/TE with heat sink	The most effective structure is PV/TE with a heat sink which dissipates heat by cooling the whole cell by ~8 ˚C. Moreover, the TE module itself shows a cooling performance of approximately 27 %	———
Ershuai Yin et al. [208]	Exp.	Passive	CPV/TE with PCM	The temperature of the CPV-PCM-TE hybrid system can be reduced by about 30 ˚C compared to the CPV-TE hybrid system	The output power of the PV-PCM-TE hybrid system increased by 23.52 % compared to the CPV-TE hybrid system.
Benghanem et al. [23]	Exp.	Active	Thermoelectric (TE) as a cooling module in hot climatic conditions	———	The efficiency of solar cells decreases with an increase of their temperature by 0.5 % for each ˚C increase in temperature.
Fatemeh Rajaee et al. [209]	Exp.	Active	PV/TE with pure and combined PCM (with Alumina powder), and cobalt oxide nanofluid	———	The use of both PCM and 1 % nanofluid enhanced the overall electrical efficiency by 4.52 %.
Hamidreza Najafi et al. [210]	Num.	Active	CPV-thermoelectric Module.	This technique can successfully keep the temperature of the photovoltaic cell at a low level with a reasonable amount of electricity.	———
Yongliang Li et al. [211]	Theor.	Active	TE module to harvest solar energy from a wide spectral range along with supercooling the photovoltaic cells	———	Output power can be improved to 30 % by adding a high-quality cold energy storage system.
Machrafi [212]	Num.	Passive	PV/TE	———	The overall efficiency was increased by up to an additional 10 % (anincrease of almost 50 %)
Shohreh Soltani et al. [213]	Exp.	Active	hybrid photo-voltaic/thermoelectric system with nanofluids	———	SiO2/water nanofluid cooling showed the best improvements in power production and efficiency with 54.29 % and 3.35 %, respectively.
Rouhollah Salehi et al. [214]	Exp.	Passive	PV/TE with an anodized heat sink	Temperature can be reduced by an average of 10 ˚C	The efficiency and power increased by an average of 10.50 % and 10.50 %, respectively.
Tengfei Cui et al. [215]	Exp.	Active	CPV/TE with PCM and water cooling	———	The efficiency of the PV-PCM-TE hybrid system with water cooling was higher than that of air cooling in the same working environment.
Alireza Akbar et al. [216]	Exp.	Active	Hybrid photovoltaic-thermal-thermoelectric with nanofluid	The temperature of PVT-TE was decreased by 1.77 ˚C, 9.76 ˚C, and 13.17 ˚C using water, SiO2-H2O, and Ag-H2O, respectively	Efficiency improved by 13.09 %, 16.17 %, and 20.68 % by using water, SiO2-H2O, and Ag-H2O respectively.

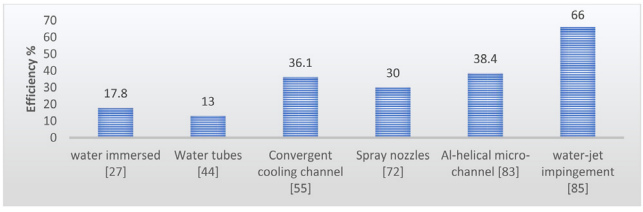


Fig. 28. The highest efficiency improvement for water cooling systems reported in the literature.

been tested in different conditions. These cooling techniques can enhance electrical output efficiency, with improvements ranging from 4 % to 29 % (as shown in Fig. 29). The most effective method observed was the combination of a geothermal heat exchanger and cooling ducts. While the cost of setting up a geothermal system can be high due to its complexity, it is a viable option for places with high temperatures. In regions with more moderate temperatures, the use of a simple finned heat sink is effective in cooling photovoltaic cells. However, it is important to note that this approach adds additional weight to the panels. Another approach, known as forced convection, is effective in cooling, although it demands considerable fan power. According to Hallal et al. [113],

Table 13
Summary of most studies conducted on photovoltaic panels with heat pipes as passive cooling techniques.

Authors	Study type	Cooling method	Thermal results	Electrical results
Hossein Alizadeh et al. [217]	Num.	Single-turn pulsating heat pipe PHP	———	Electrical power improved by 18 %
Meysam Moradgholi et al. [218]	Exp.	PV/T with heat pipe (thermosyphon type)	With methanol, the thermal efficiency was improved by 16.35 %, while acetone improved by 45.14 %	With methanol, the average electrical power was improved by 5.67 %, while acetone improved by 7.7 %
Shuang-Ying Wu et al. [219]	Theor.	Heat pipe photovoltaic/thermal (PV/T)	The overall thermal was 63.65 %	The overall electrical, and exergy efficiencies were 8.45 %, and 10.26 %, respectively
Jouhara et al. [220]	Exp.	PV/T with flat heat pipes (heat mat)	The thermal efficiency of the PV/T mat was 50 %	The electrical output of PV/T increased by about 15 %
Adham Makki et al. [221]	Num.	Heat pipe-based photovoltaic-thermoelectric generator (HP-PV/TEG)	———	Integrating TEG modules with photovoltaic cells in this way helps to improve the performance of photovoltaic cells
Meysam Moradgholi et al. [222]	Exp.	Nanofluid in a two-phase closed thermosyphon (TPCT)	Reduce PV/T temperature to 14.52 ˚C	Generate 1.42 W more power, and the average electrical energy increased by 1 %
Sandeep Koundinya et al. [223]	Num. & Exp.	Finned heat pipe	Showed a maximum temperature decrease of 13.8 K using this method.	———
Yiping Wang et al. [224]	Exp.	Atmospheric plate thermosyphon APT	The APT cooling system can effectively reduce the temperature of the photovoltaic cells.	———
Hossein Alizadeh et al. [225]	Num.	Flat plate pulsating heat pipe	The CLPHP-based increases the improvement rate by 23 %	———

Table 14
Summary of most experimental studies conducted on photovoltaic panels integrated with heat pumps as active cooling techniques.

Authors	Cooling method	Thermal results	Electrical results
Hongbing Chen et al. [227]	PV panels as evaporators with a small heat pump	———	The electrical efficiency of the PV panel was improved by up to 1.9 %
Qu Minglu et al. [228]	PV/T integrated with dual-source heat pump (DSHP)	The operating temperature of the PV/T plate could be reduced up to 45 ˚C	The electrical conversion efficiency yielded a surplus of 10.3 %.
Jiannan Tong et al. [229]	Water-cooled PV/T collector combined with a dual-source heat pump	———	The maximum generation efficiency was up to 15.3 %, and the maximum generated power was 827 W,

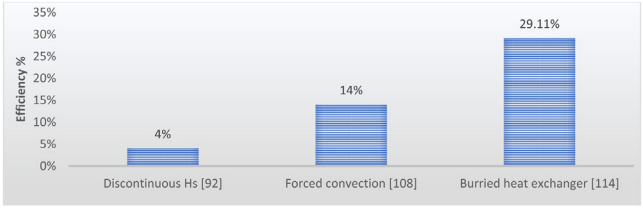


Fig. 29. The highest efficiency improvement for air cooling systems reported in the literature.

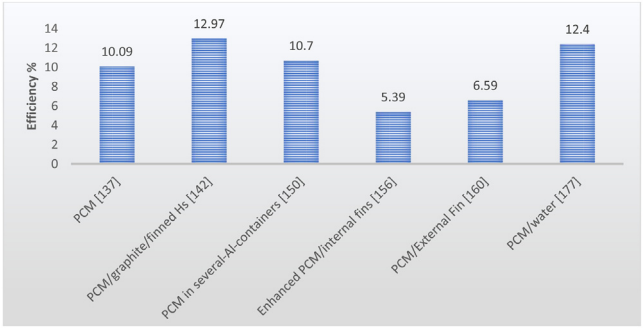


Fig. 30. The highest efficiency improvement for PCM cooling systems reported in the literature.

the maximum permissible temperature is 45 ˚C to optimize energy production. furthermore, if thermal energy is required, cooling ducts can be employed to extract heat. However, its efficiency depends on the required temperature.

The study of PCM cooling techniques in the literature showed that the most significant improvement in electrical efficiency was in the range of 5 % to 13 % (as shown in Fig. 30). Notably, using a composite PCM with a finned heat sink achieved the highest efficiency. However, The main drawback of this method is its low thermal conductivity and volume change when the PCM melts, which in turn leads to poor temperature management. Researchers have tried different ways to fix this, like mixing PCM with nanoparticles, adding a finned heat sink to the PCM, developing shape-stable PCM, and many other suggestions. Despite these challenges, the simple design, low cost, and durability of this cooling system make it a very promising technique.

The literature introduces several additional cooling techniques for photovoltaic systems, including liquid immersion, employing nanofluids as coolants, radiative cooling, thermoelectric cooling, heat pipes, and heat pumps. These approaches contribute to enhancing photovoltaic panel performance, resulting in improvements ranging from 10 % to 90 % (as illustrated in Fig. 31). Notably, the most substantial improvement was observed through the utilization of water-nanofluid with nano-PCM. The incorporation of nanofluids at varying concentrations enhances the thermal properties of water, leading to a significant increase in electrical efficiency. However, extracting heat from the wa-

Table 15
Summary of most studies conducted on photovoltaic panels with other uncategorized cooling methods.

Authors	Study type	Cooling Type	Cooling method	Thermal results	Electrical results
Ruobing Liang, Jili Zhang et al. [230]	Exp.	Active	Graphite-filled hybrid PV/T solar collector	———	The highest electrical efficiency was 7.2 %
Ghadiri et al. [231]	Exp.	Active	PVT with water and ferrofluid under constant and alternating magnetic field	———	The overall efficiency increased to about 50 % at 50hz
Rostami et al. [232]	Exp.	Active	Using high-frequency ultrasound and nanofluid	The temperature was reduced by 57.25 %	The maximum power increase was 51.1 %
Heidari et al. [233]	Exp.	Active	Ferrofluid and a rotating magnetic field	Thermal efficiency increased by 30 %.	The maximum increase in power was 47.5 %
Sudhakar et al. [234]	Exp.	Active	Encapsulated phase change material (HS29) filled into cylindrical thermal energy storage (CTES) tank	The average differences in maximum and minimum temperatures between ambient air and air entering the PV collector were 5.4 °C and 3.4 °C, respectively.	———
Wang et al. [235]	Exp.	Active	Active phase-change cooling technique based on porous media	———	The maximum power improvement was 19.32 %.
Ahmad El Mais et al. [236]	Theor.	Active	Flipping technique	This technique could keep the PV temperature at a specified low value	———

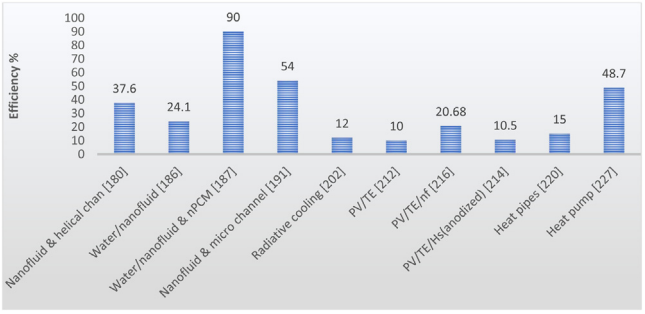


Fig. 31. The highest efficiency improvement for other cooling systems reported in the literature.

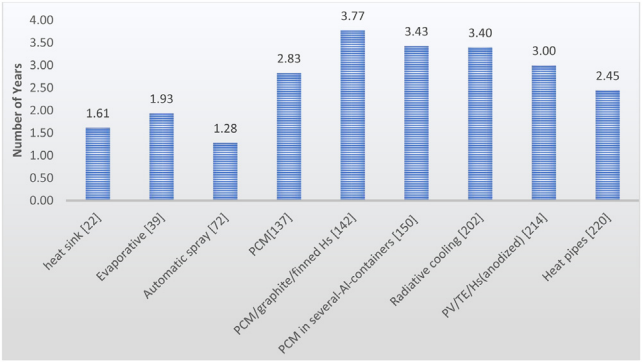


Fig. 32. The payback period for various cooling techniques found in this literature.

ter for practical usage necessitates an external heat exchanger, thereby introducing an additional cost. Using thermoelectric modules for photovoltaic cooling was observed to have a minimal impact due to the limited temperature difference, whereas the addition of heat sinks combined with forced-air cooling showed significant cooling of the photovoltaic cells. Additionally, the implementation of radiative cooling was found to be uncomplicated and applicable to all installed PV systems by incorporating a passive cooling optical filter, resulting in a moderate cooling effect.

For a more accurate comparison, the payback period is calculated for various cooling techniques based on the additional initial cost of each technique and their net electrical energy yield, taking into account the energy added or consumed by the cooling methods. However, this study does not incorporate the consideration of thermal energy.

This chart (Fig. 32) presents a rating of different cooling techniques based on their respective payback periods in years. A shorter payback period indicates a faster return on investment and greater cost-effectiveness. It's worth noting that automatic spray cooling stands out with the shortest payback period of 1.279 years, indicating its potential for rapid cost recovery and efficiency. Heat sinks also offer a relatively short payback period of 1.614 years, making them an economically attractive option for thermal management. Evaporative cooling displays a payback period of 1.935 years, demonstrating its competitive performance in terms of cost-effectiveness. In contrast, PCM, PCM/graphite/finned heat sinks, and PCM in multiple aluminum enclosures exhibit slightly longer recovery times (2.828, 3.772, and 3.429 years, respectively), Underlining their effectiveness but necessitating a longer-term investment perspective. Similarly, radiative cooling, PV/TE/Hs (anodized), and heat pipes reveal payback periods of 3.397, 3.000, and 2.446 years, respectively. Lastly, nanofluid-based cooling methods (nanofluids with cooling channels and PV/TE/nanofluids) demonstrate relatively long payback periods of 3.816 and 3.878 years, underscoring the importance of considering not only their electrical performance but also their financial implications when selecting a suitable cooling technology for a specific application.

Table 16 The table summarizes various cooling techniques for photovoltaic panels. Each technique is briefly outlined with its key advantages and disadvantages, providing an overview of options to enhance panel performance.

In this section, we explored various cooling techniques for photovoltaic panels and their impact on electrical efficiency. The primary objective was to identify promising techniques that warrant further research. Water cooling methods, including water spraying, immersion, circulation through tubes, jet impingement, and evaporative cooling, show efficiency gains of 13 % to 66 %. While initial setup costs are involved, the effectiveness in harsh environments and the potential to reuse extracted heat make them compelling. Air cooling methods, such as heat sinks, fans, ducts, and finned plates, enhance electrical output by 4 % to 29 %. Geothermal heat exchangers and cooling ducts offer effective cooling in high-temperature regions. Phase change materials (PCMs) present efficiency improvements of 5 % to 13 %, with composite PCMs and finned heat sinks showing promise. Additional techniques, like liquid immersion, nanofluids, radiative cooling, thermoelec-

Table 16
Advantages and disadvantages of different cooling techniques.

COOLING TECHNIQUES	ADVANTAGES	DISADVANTAGES
WATER IMMERSED	• High efficiency • Low cost • Maintain good temperature distribution	• Performance depends on the depth of immersion • Possibilities of soil participation. • Cannot be applied to the already installed PV
NATURAL WATER CIRCULATION	• Hot water can be reused • No need for pump power • Moderate efficiency	• Complex system
PHOTOVOLTAIC THERMAL	• Increased combined thermal and electrical efficiency • Extracted heat can be reused	• High cost • Needs for pumping power • Consumed much power • Needs for regular maintenance
EARTH-WATER HEAT EXCHANGER	• High efficiency • Can maintain PV at a low temperature. • Closed system (saving water)	• High cost • Complex system • Needs for pumping power. • Consumed much power • Needs for regular maintenance
SPRAY NOZZLES	• High efficiency • Uniform Cooling • Dual role (cooling & cleaning) • Hot water can be reused	• High cost • Needs for pumping Power • Consumed much Power • Needs a water filtration system
MICROCHANNEL COOLING	• High efficiency • Uniform temperature distribution • Can be used in conjunction with another cooling technique	• High cost • Required pumping power • Designed for cooling CPV • Consumed much Power
JET IMPINGEMENT	• High efficiency • Can extract a large amount of heat	• High cost • Complex system • Difficult integration with PV
HEAT SINK	• Good efficiency • Uniform cooling • Can be applied to most already installed PV • Can be Applied with free/forced convection • No need for maintenance	• Extracted heat is wasted • Difficult fabrication • Moderate cost • Its performance is related to the ambient conditions
AIR COLLECTOR	• Good efficiency • Low cost • Simple system • Hot air can be reused • Can be Applied with free/forced convection	• Non-uniform temperature distribution • Needs for fans, or suction devices • Consumed power • Needs for maintenance
PCM	• Good efficiency • Low cost • Uniform temperature distribution • High latent heat • Can be applied with multiple cooling techniques • Long lifetime	• Low thermal conductivity. • Less efficient in cold climatic conditions • Risk of leakage • Thermal performance degrades over time • The heat extracted is wasted
NANOFLUID	• Low cost • Good thermal property • Adding nanofluid to water or PCM, significantly improved the efficiency	• The external heat exchanger must be added when used with water • Risk of sedimentations (needs to mix regularly).
EVAPORATIVE	• Passive cooling technique • Good efficiency	• Extracted heat is wasted.
LIQUID IMMERSION	• Low cost • Uniform temperature distribution • Reduce reflection	• Performance depends on the depth of immersion • Ionized liquid reduces efficiency due to its electrical conductivity
THERMOELECTRIC	• Good efficiency • Produce extra electric power • Can be applied with most cooling techniques. • Uniform temperature distribution	• Needs high-temperature difference • Limited for CPV • Extracted heat is wasted
HEAT PUMP	• High efficiency • Uniform temperature distribution	• High cost • Complex system • Needs for regular maintenance

tric cooling, heat pipes, and heat pumps, contribute to improvements of 10 % to 90 %. Water nanofluids with nano-PCM stand out as highly efficient. Lastly, a chart rates these techniques based on their respective payback periods in years, emphasizing the financial implications of cooling technology selection for specific applications. Additionally, a table summarizes the advantages and disadvantages of these cooling methods, providing an overview of options for enhancing panel performance while considering the payback period.

6. Conclusions

This study provides a general and detailed review of advanced cooling techniques for improving the performance of photovoltaic panels. These techniques consist of several methodologies, which are reviewed and discussed to focus on promising cooling techniques for further study. Moreover, this review introduces a new classification of research papers, which categorizes all similar techniques with illustrations, tables, and graphs from different perspectives and classifications, and provides an updated research study of photovoltaics using different cooling techniques.

The literature reported that the highest improvement was achieved when using water cooling techniques, with an increase in electrical efficiency ranging from 6 % to 82.6 % achieved using different types of cooling techniques. Among them is the water spray cooling system, which is very effective in dusty locations and can maintain the PV surface temperature at low limits with a clean surface. In addition, evaporative cooling techniques improve the electrical efficiency by approximately 7.6 %–23 % without using pumping power. Moreover, adding nanofluids to water at different concentrations significantly improved the electrical efficiency by up to 24 %. In addition, air cooling methods were found to have a moderate effect on increasing efficiency, ranging from 4 % to 29 %, using different cooling techniques. On the other hand, phase change materials (PCM) have the least effect on improving electrical efficiency, with a maximum improvement between 5 % and 13 % among different types of PCM. Using nano-enhanced PCM with water-nanofluid can result in up to 93 % improvement in electrical efficiency. Radiative cooling techniques are advanced techniques that aim to enhance and filter the spectral wavelength to remove infrared and ultraviolet rays by keeping visible light unaffected. This method is economical and can improve electrical efficiency by more than 4.5 %. In addition, thermoelectric cooling techniques are always used in conjunction with various cooling technologies, such as fin heat sinks, water cooling, PCM, water nanofluids, fans, and many other techniques. The combination of a PV/TE system with various cooling techniques can improve electrical efficiency by approximately 3.35 % to 10.5 %. Furthermore, automatic spray cooling has the shortest payback period at 1279 years, closely followed by heat sinks at 1614 years. Evaporative cooling decreases at 1935 years. Conversely, technologies such as phase change materials have varying payback periods between 2828 and 3772 years, necessitating a broader investment horizon due to slightly longer payback times. In addition, radiative cooling, PV/TE/Hs (oxidized), and heat pipes extend from 2446 to 3397 years, while nanofluidic-based methods cover a range of 3816 to 3878 years. The choice of cooling technique must balance electrical performance with financial implications.

In conclusion, this comprehensive analysis confirms the significance of advanced cooling techniques in optimizing the performance of photovoltaic panels. By evaluating a range of methodologies and presenting a novel classification framework, this review identifies key strategies for further exploration. The diverse approaches, from water cooling to radiative techniques and thermoelectric integration, all contribute to advancing the efficiency of photovoltaic systems. As research continues in this dynamic field, the findings presented here offer valuable insights for shaping the future of photovoltaic cooling technologies.

Future recommendations

Going forward, research in the field of photovoltaic cooling could focus on:

1. Comparative Analysis: Undertake comprehensive comparative evaluations of cooling techniques to distinguish their benefits in diverse scenarios.
2. Material Advancements: Explore novel materials that enhance thermal conductivity and phase change properties for better cooling efficiency
3. Efficiency and maintenance balance: Developing intelligent control systems that strike a balance between maintaining high efficiency and reducing energy consumption.
4. Long-term Performance: Focus on studying the long-term durability and effectiveness of cooling solutions under different environmental scenarios.
5. Economic Feasibility: Examine the economic viability of implementing these techniques across different scales.
6. Environmental Impact: Assess the environmental implications of different cooling methods through lifecycle analyses.

Additionally, conducting an experimental setup study that incorporates PV panels equipped with an automatic spray cooling system, PV panels with heat sinks, PV panels with evaporative techniques, and standard PV panels would facilitate a comprehensive comparison of these passive cooling techniques under consistent weather conditions.

Furthermore, exploring alternative setups that integrate PV panels with evaporative techniques and heat sinks, or combine PV panels with sprayer systems and heat sinks, and comparing them to standard PV panels, would provide a more thorough assessment of their collective efficiency and effectiveness. These studies could offer valuable insights into the synergies of combining different cooling methods to enhance photovoltaic performance.

Moreover, it's a valuable suggestion to explore scenarios where hot water is required for other purposes. In these cases, conducting experiments with a standard PV panel and PV panels integrated with either pure or nano-enhanced phase change materials and water in various configurations, incorporating different PCMs with varying melting points and thicknesses, would provide a robust basis for precise comparisons between these techniques. This comprehensive study would not only shed light on the most effective combinations and configurations for harnessing both electrical and thermal energy from PV systems but also highlight the potential for dual-purpose applications of solar energy.

By addressing these specific areas, researchers can drive advances in photovoltaic cooling methods, and contribute to more efficient and sustainable energy solutions.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Mohamad Abou Akrouch: Data curation, Formal analysis, Investigation, Methodology, Validation, Writing – original draft. **Khaled Chahine:** Formal analysis, Investigation, Writing – review & editing. **Jalal Faraj:** Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Supervision, Writing – review & editing. **Farouk Hachem:** Formal analysis, Investigation, Writing – review & editing. **Cathy Castelain:** Formal analysis, Investigation, Writing – review & editing. **Mahmoud Khaled:** Conceptualization, Formal analysis,

Investigation, Methodology, Project administration, Supervision, Writing – review & editing.

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